Computer Vision and Image Understanding 251 (2025) 104253

Contents lists available at ScienceDirect

Computer Vision and Image Understanding

journal homepage: www.elsevier.com/locate/cviu

Building extraction from remote sensing images with deep learning: A survey on vision techniques

Yuan Yuan *, Xiaofeng Shi, Junyu Gao *

School of Artificial Intelligence, OPtics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China

A R T I C L E I N F O

Communicated by Juergen Gall

Keywords:
Building extraction
Deep learning
Review
Semantic segmentation
Polygon generation

A B S T R A C T

Building extraction from remote sensing images is a hot topic in the fields of computer vision and remote sensing. In recent years, driven by deep learning, the accuracy of building extraction has been improved significantly. This survey offers a review of recent deep learning-based building extraction methods, systematically covering concepts like representation learning, efficient data utilization, multi-source fusion, and polygonal outputs, which have been rarely addressed in previous surveys comprehensively, thereby complementing existing research. Specifically, we first briefly introduce the relevant preliminaries and the challenges of building extraction with deep learning. Then we construct a systematic and instructive taxonomy from two perspectives: (1) representation and learning-oriented perspective and (2) input and output-oriented perspective. With this taxonomy, the recent building extraction methods are summarized. Furthermore, we introduce the key attributes of extensive publicly available benchmark datasets, the performance of some state-of-the-art models and the free-available products. Finally, we prospect the future research directions from three aspects.

1. Introduction

1.1. Background and literature trends

Due to diverse modality and wide viewing range, remote sensing data is rich in valuable information (Cheng et al., 2020; Keisler et al., 2019). Besides, with the continuous development of observation technology, the quality and quantity of remote sensing data have been significantly improved, bringing opportunities to better capture the interested land cover objects (Jing et al., 2023; Guo et al., 2024). As one of the most important spaces for human activities, the building is always concerned in remote sensing images. Concretely, building footprints information is vital for a range of geospatial applications, such as urban planning (Caye Daudt et al., 2019; Liu et al., 2022c), population estimation (Liu et al., 2021a; Boo et al., 2022), disaster emergency (Liu et al., 2018; Xie et al., 2022) and 3D city reconstruction (Li et al., 2021a; Wang et al., 2021b). Consequently, building extraction from remote sensing data, aiming to distinguish building and non-building parts automatically, has become a hot topic in both computer vision and remote sensing communities.

Early building extraction approaches leverage prior knowledge and shallow features (Sirmacek and Unsalan, 2009; Xia et al., 2019; Awrangjeb et al., 2013; Li et al., 2015; Huang et al., 2016). For example, Sirmacek and Unsalan (2009) use *Scale Invariant Feature Transform* (SIFT) to detect the building key points where the multiple subgraph matching and graph cut methods are then employed. Xia et al. (2019) extract buildings from very high-spatial-resolution remote sensing images based on geometric salience. However, due to factors such as the multi-scale features of buildings and complex backgrounds in images, these methods face limitations in accuracy and adaptability to complex backgrounds. In recent years, *Deep Learning* (DL), especially *Convolutional Neural Networks* (CNNs), has made breakthrough progress in the field of computer vision (Krizhevsky et al., 2012; He et al., 2016; Ronneberger et al., 2015; Shelhamer et al., 2017). In general, building extraction can be regarded as a variation of image segmentation. Inspired by this, DL-based building extraction methods begin to emerge. For example, Maggiori et al. (2017b) introduce the *Fully Convolutional Networks* (FCN) architecture and trained it by a two-step approach using a mass of inaccurate labels and small amounts of manual labels. Prathap and Afanasyev (2018) integrate RGB, pansharpened-multispectral, and *Geographic Information System* (GIS) data by the ensemble of three improved models to detect building footprints. Since it can learn generalized representations without manual design, DL is better suited for handling complex scenes. The general workflow of these methods is illustrated in Fig. 1.

* Corresponding authors.
E-mail addresses: y.yuan@nwpu.edu.cn (Y. Yuan), sxf118@foxmail.com (X. Shi), gjy3035@gmail.com (J. Gao).

https://doi.org/10.1016/j.cviu.2024.104253
Received 26 November 2023; Received in revised form 19 October 2024; Accepted 1 December 2024
Available online 12 December 2024

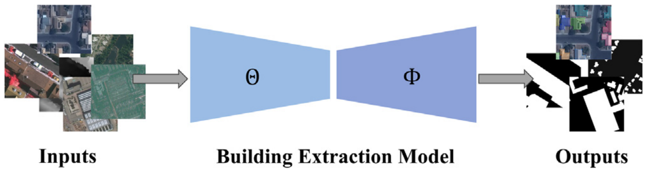

Fig. 1. Illustration of building extraction. The left is the input remote sensing images, which may encompass multiple modalities such as optical and radar images. The right is the extracted building regions, generally referring to building footprints or rooftops. The output can be represented in two forms: masks and polygons. The mask, as shown in the black-and-white image on the left, uses white to indicate areas predicted as buildings and black for non-building regions. The polygon, depicted in the colored image on the left, is typically represented by the key corner points of the buildings.

This study aims to provide a comprehensive survey of recent DL-based building extraction methods. To ensure the survey's comprehensiveness, representativeness, and timeliness, literature searches were conducted primarily on mainstream academic platforms such as Google Scholar and IEEE Xplore, using keywords like "deep learning", "remote sensing imagery", "building extraction", and "building segmentation". From the initial search results, papers published after 2018 in peer-reviewed journals and conferences within the fields of computer vision or remote sensing were primarily retained. Relevance to the review's theme and technical innovation was assessed through titles, abstracts, and keywords. An brief overview of the original search results is shown in Fig. 2, and the overall situation of the constructed literature database is presented in Fig. 3 (both figures exclude data from 2024, as the ongoing nature of 2024 research at the time of this survey prevents a complete trends). From Fig. 2, it is evident that the number of publications has increased annually, with representative methods predominantly employing DL since 2018. Fig. 3 illustrates that the final literature database comprises over 300 papers, with the highest number of publications in 2022 and 2023. Annually, journal articles constitute the majority, and studies focusing on non-radar data and multi-source data fusion prevail. Based on these observations, the following trends can be identified: (1) The building extraction task has attracted increasing attention from researchers; (2) The transition from traditional methods to DL has been completed in recent years, establishing DL as the mainstream approach; (3) Although existing literature predominantly focuses on optical data, research involving non-optical data and multi-source data fusion is on the rise.

1.2. Existing surveys

Building extraction involves many fields such as image segmentation, deep learning, remote sensing image understanding, and multi-source data fusion. Although reviews on these topics (Yu et al., 2018; Zang et al., 2021; Yuan et al., 2021; Ghamisi et al., 2019) can provide illuminating perspectives to grasp research trends, they do not focus on the building extraction task. There are several reviews of building extraction published in the past decades (Mayer, 1999; Tomljenovic et al., 2015; Ghanea et al., 2016). However, they concentrate on traditional methods rather than recent DL-based methods.

To the best of our knowledge, there are four systematic surveys of building extraction with DL (Shaloni et al., 2020; Luo et al., 2021; Li et al., 2022a, 2024c), as summarized in Table 1. These surveys have laid a solid foundation for subsequent research and driven technological advancement. Nevertheless, as research has progressed, they face certain limitations in reflecting the latest developments and trends in DL-based building extraction methods. The review by Shaloni et al. (2020), being an earlier work, covers DL methods but lacks recent progress. The study by Luo et al. (2021) focuses on network architectures but overlooks key aspects such as data volume, data modality, resolution, and polygonal output requirements. Li et al. (2022a) and Li et al. (2024c) do not fully concentrate on DL methods and discuss a range of tasks related to building extraction. Moreover, all four surveys either pay limited attention to radar data or restrict their discussion of multi-source data fusion to optical-LiDAR fusion, without comprehensively summarizing the current state and trends of multi-source data in building extraction.

As previously mentioned, DL methods have become mainstream in building extraction, and research on non-optical data and multi-source data fusion is increasingly prevalent. Therefore, there is a need for a survey that comprehensively focuses on the application of DL in building extraction and systematically covers non-optical data, multi-source data fusion, and polygonal output requirements.

1.3. Objective of this survey

This survey aims to provide a comprehensive and in-depth analysis on recent DL-based building extraction methods, systematically covering the full process of DL from both internal and external model perspectives, including representation, learning strategies, data types, and model outputs. Specifically, the systematic review of learning strategies, radar data, multi-source data fusion, and polygonal outputs serves as a significant complement to existing surveys.

To achieve this goal, the following work are conducted:

① Novel Taxonomy. Recent research is organized from both internal and external model perspectives, covering four key aspects. This taxonomy aligns with the technical paradigms of DL and the challenges in building extraction (see Section 3), providing researchers with a clear and comprehensible analytical framework.

② Comprehensive Review. Over 200 papers are systematically and thoroughly reviewed. The motivations and core ideas of representative methods are described. An in-depth discussion of these methods is also provided.

③ Extensive Resources. Abundant public datasets, evaluation metrics, some *State-Of-The-Art* (SOTA) methods performance on benchmark datasets, and freely available building footprint products are compiled and summarized, providing convenience to the research community.

④ Future Direction. Based on a comprehensive review of the existing literature and the latest developments in the field, this survey delves into the development trends of building extraction from three perspectives, offering insights for future research directions.

The remainder of this paper is organized as presented in Fig. 4. Section 2 makes a brief introduction of the preliminaries of building extraction methods. Section 3 summarizes the challenges of building extraction using DL from both internal and external model perspectives and introduces a novel taxonomy. The recent advanced building extraction methods are discussed in Sections 4 and 5. These methods are organized according to the proposed taxonomy and are arranged chronologically in each sub-category to better show the development venation. Section 6 introduces the publicly available resources and discusses the performance of some advanced building extraction approaches. The outlook of future direction is presented in Section 7. And Section 8 concludes this article.

2. Preliminaries

In this section, various types of deep semantic segmentation models and the early DL-based building extraction are briefly recapped, in order to provide an entry-level guidance.

2.1. Deep learning-based segmentation

While object detection methods can be applied to extract buildings from remote sensing imagery, they only provide the location and rectangular bounding box of the buildings. This falls short of meeting the requirements for mapping or environmental detection effectively. Additionally, in scenes with densely distributed small buildings, object detection methods are highly prone to missing detections.

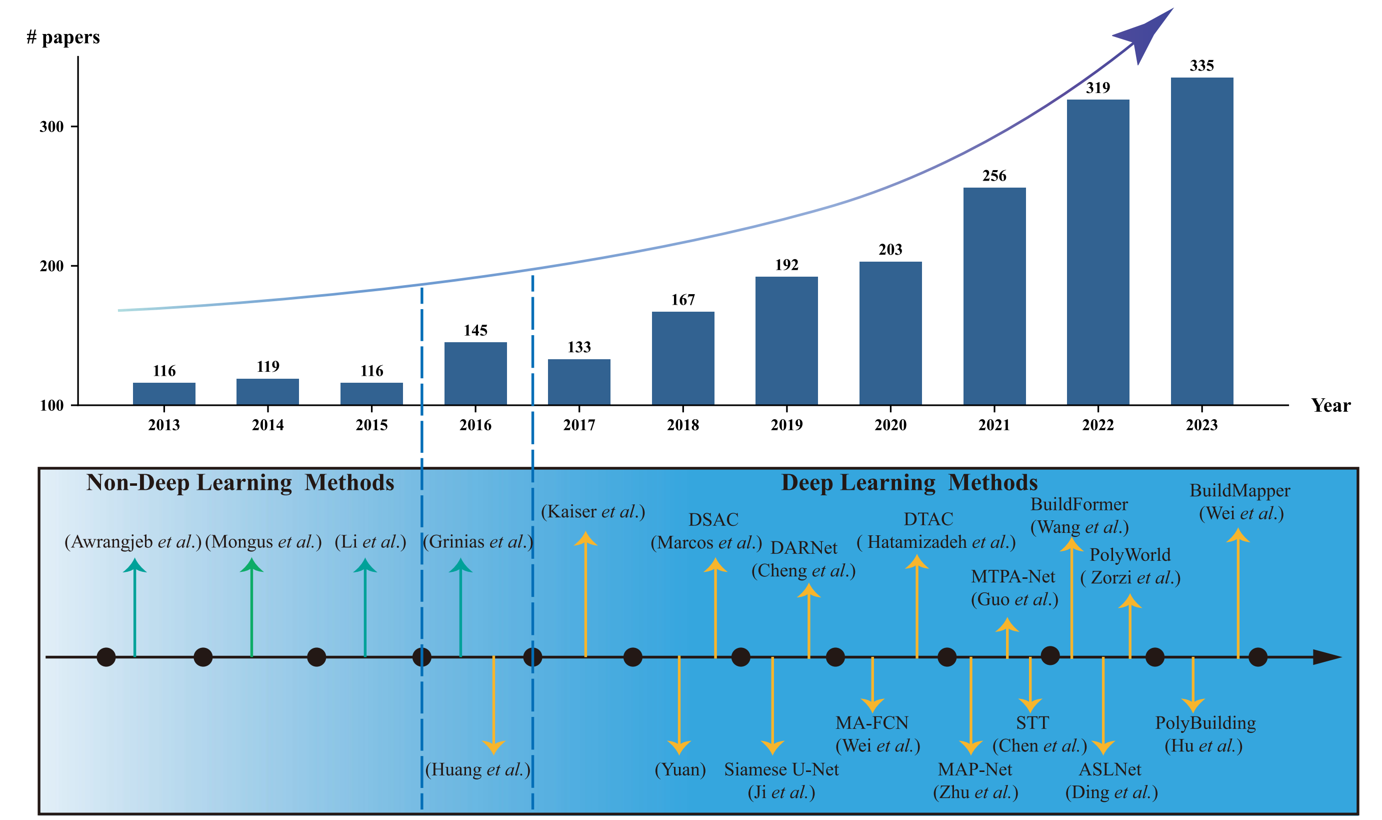

Fig. 2. A brief overview of the development of building extraction over the past decade. The top half of the figure is the statistic for published papers related to building extraction from 2013 to 2023 according to google scholar advanced search: allintitle: building (extraction OR segmentation), access on 04, October, 2024. The bottom half of the figure is a brief chronology of building extraction. Representative methods (highly cited papers or those published in top-tier journals and conferences) in this figure: Awrangjeb et al. (2013), Mongus et al. (2014), Li et al. (2015), Grinias et al. (2016),Huang et al. (2016), Yuan (2018), DSAC (Marcos et al., 2018), Siamese U-Net (Ji et al., 2019), DARNet (Cheng et al., 2019), MA-FCN (Wei et al., 2020), DTAC (Hatamizadeh et al., 2020), MAP-Net (Zhu et al., 2021), MTPA-Net (Guo et al., 2021a), STT (Chen et al., 2021), BuildFormer (Wang et al., 2022a), ASLNet (Ding et al., 2022), PolyWorld (Zorzi et al., 2022), PolyBuilding (Hu et al., 2023a) and BuildMapper (Wei et al., 2023).

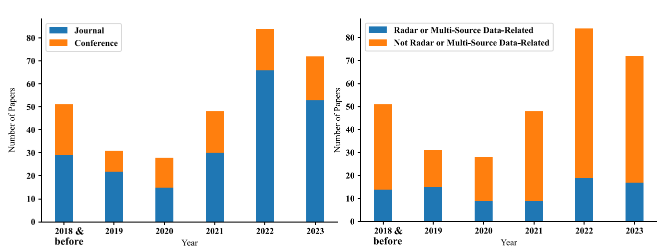

Fig. 3. The overall situation of the constructed literature database. (a) Number of journal and conference papers in the constructed database. (b) Number of papers involving radar or multi-source data in the constructed database.

Consequently, the majority of current research treats building extraction as an image segmentation task. Within deep learning-based image segmentation, there are two primary categories: semantic segmentation and instance segmentation. Semantic segmentation requires that every pixel in the image is assigned a category, which can also be regarded as a pixel-by-pixel classification task. Instance segmentation (He et al., 2017; Wang et al., 2020; Ke et al., 2022) aims to segment individual objects and assign a unique label to each of them. Generally, the design of models for instance segmentation is often more intricate compared to semantic segmentation. Moreover, for the task of building extraction, imposing constraints at the segmentation setting is not necessary to obtaining instance information (discussions in Section 4). Hence, apart from Mask R-CNN (He et al., 2017), the majority of current methods employed for building extraction rely on semantic segmentation models, as outlined in the following description.

2.1.1. CNNs-based semantic segmentation

Convolution is an operation used for extracting local spatial patterns by sliding a convolutional kernel over the input tensor to compute a weighted sum. Its core formula is:

$$y_{i,j} = \sum_{m} \sum_{n} x_{i+m,j+n} \cdot w_{m,n}, \tag{1}$$

where x is the input tensors, w is the kernel, and y is the output feature maps. Typical CNNs (Krizhevsky et al., 2012; Szegedy et al., 2015; Simonyan and Zisserman, 2015; He et al., 2016; Hu et al., 2018) are locally connected and weight-shared, consisting of convolutional layers, pooling layers, and fully connected layers. Early semantic segmentation methods use typical CNNs in a patch-wise way (Ciresan et al., 2012; Farabet et al., 2013), resulting in some issues like computational redundancy and limitation of input size.

CNN-based semantic segmentation models typically employ an encoder–decoder architecture (as shown in Fig. 5(a)), where the encoder progressively extracts high-level features through convolution and pooling operations. The decoder gradually restores spatial resolution using either simple upsampling or carefully designed modules to produce pixel-level results. FCN (Shelhamer et al., 2017) is an epoch-making work of semantic segmentation for overcoming the above issues by using only convolutional layers. Furthermore, SegNet (Badrinarayanan et al., 2017) adds batch normalization operation to improve training efficiency. *U-shape Network* (U-Net) (Ronneberger et al., 2015) concatenates the features extracted from the encoder to the symmetric part from the decoder, achieving the transmission of multi-level features at different scales. DeppLab series methods are representative works that can leverage multi-scale features and spatial context well (Chen et al., 2014, 2018a, 2017, 2018b). Among them,

Table 1
Comparison with existing surveys.

Surveys	"Building Extraction from Remote Sensing Images: A Survey" (Shaloni et al., 2020)	"Deep Learning-Based Building Extraction from Remote Sensing Images: A Comprehensive Review" (Luo et al., 2021)	"A Review of Building Detection from Very High Resolution Optical Remote Sensing Images" (Li et al., 2022a)	"A Review of Building Extraction From Remote Sensing Imagery: Geometrical Structures and Semantic Attributes" (Li et al., 2024c)	"Building Extraction from Remote Sensing Images with Deep Learning: A Survey on Vision Techniques" (Ours)
Year & Pub.	2020 ICACCCN	2021 Energies	2022 GIScience & Remote Sensing	2024 IEEE Transactions on Geoscience and Remote Sensing	2025 Computer Vision and Image Understanding
Existing surveys are reviewed	No	No	No	Briefly	Comprehensively
Reviewed task type	Footprint extraction	Footprint extraction	Footprint extraction, height retrieval, type classification, change detection	Footprint extraction, height retrieval, type classification, change detection, façade/roof segmentation, annotation correction	Footprint extraction
Number of referenced papers	17	129	143	182	259
Types of techniques covered	Traditional segmentation, classical machine learning, vanilla deep learning model	Deep learning	Physical rule, traditional segmentation, classical machine learning, deep learning	Heuristic feature design, deep learning	Deep learning
Modality	Optical, LiDAR	Optical, LiDAR	Optical, LiDAR	Optical, LiDAR, SAR	Optical, LiDAR, SAR
Taxonomy	Based on technique types	Based on network architectures	Based on technique types	Based on task types	Based on model-internal and model-external
Number of building extraction datasets summarized	4	3	22	17	29
Multi-source fusion	Optical-LiDAR	Optical-LiDAR	Optical-LiDAR	Optical-LiDAR	Optical-Optical, Optical-LiDAR, Optical-SAR
Polygonal outputs	No	No	Yes	Yes	Yes
Review free-available productions	No	No	No	No	Yes

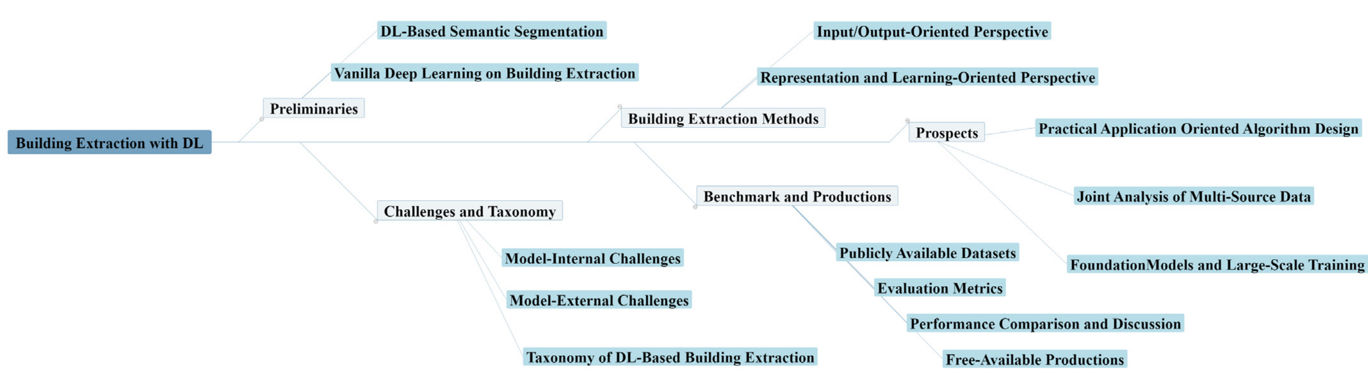

Fig. 4. The main structure of this article.

Atrous Spatial Pyramid Pooling (ASPP) module inspired by PSPNet (Zhao et al., 2017) is the most widely known. In addition, attention-based approaches (Chen et al., 2016; Fu et al., 2019) also achieve impressive results.

CNN-based semantic segmentation methods efficiently extract local features, offering high computational efficiency and segmentation performance. And the representation capability can be enhanced through network depth and structural design. However, due to the limited receptive field of convolution operations, these methods often struggle to capture long-range dependencies and global information, leading to potential weaknesses in complex scenes. Additionally, the fixed receptive field constrains the model's ability to handle objects at varying scales.

2.1.2. *Transformer-based semantic segmentation*

Transformer, a sequence-to-sequence model consisted of *Multi-Head Self-Attention* (MHSA), is originally proposed by Vaswani et al. for natural language processing (Vaswani et al., 2017). MHSA captures different feature relationships from the input sequence by computing multiple attention mechanisms in parallel. The outputs of each head

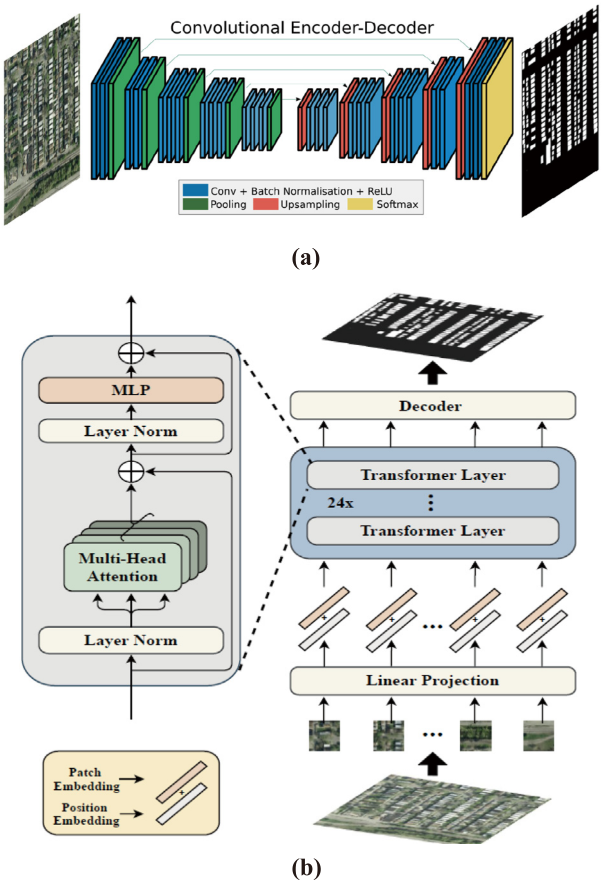

Fig. 5. Illustration of the general workflow of (a) CNN-based and (b) transformer-based semantic segmentation.
Source: Adapted from Badrinarayanan et al. (2017), Zheng et al. (2021a).

are concatenated and then linearly transformed to obtain the final representation. The core formula of attention mechanisms is as follows:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \tag{2}$$

here, the query matrix Q, the key matrix K and the value matrix V are obtained by multiplying the input features with learnable weight matrices respectively, and d_k is the dimension of the key matrix. More recently, transformer architecture has also shown great potential for vision tasks, known as *VIsion Transformer* (ViT) (Dosovitskiy et al., 2021). ViT adapts the transformer model to image processing by first splitting an input image into non-overlapping patches. Each patch is then linearly embedded into a fixed-length vector, combined with positional encodings, and fed as a token sequence into a standard Transformer encoder. In this way, ViT enables the model to capture global context through self-attention across all patches and achieves competitive performances compared to several SOTA CNN.

Encouraged by the great success of ViT, researchers attempt to use transformers for semantic segmentation, and the general workflow is illustrated in Fig. 5(b). Zheng et al. propose SETR (Zheng et al., 2021a) to apply a pure transformer as the backbone and the convolutions with two different strategies as the decoder. Strudel et al. (2021) employ transformer through the whole network as both encoder and decoder. To address the issue that the objects of interest are usually variant in scale, Liu et al. (2021b) present Swin Transformer, a hierarchical transformer with shifted windows. And Cao et al. (2021) organize blocks of Swin transformer as U-Net architecture. Furthermore, Xie et al. (2021) design a simple framework named SegFormer, which show excellent robustness and efficiency.

Transformer-based semantic segmentation methods effectively model long-range dependencies and global information, achieving superior performance in complex scenes and multi-scale objects. Nevertheless, these approaches often require significant computational resources and large datasets for training, leading to high costs when processing high-resolution remote sensing imagery. Moreover, transformer-based methods may be less effective than CNN-based methods in capturing local details.

2.2. Vanilla deep learning on building extraction

With the widespread application of DL in the field of vision, some researchers directly employ vanilla DL methods (including unmodified or only slightly optimized network) for building extraction. Alshehhi et al. (2017) propose a single patch-based CNNs to extract roads and buildings simultaneously. Kaiser et al. (2017) compare the performance of FCN trained with the supervision of open image and map data from the Internet and that trained with manually labeled pixel-accurate ground truth. They point out that it is feasible to reduce the manual annotation burden by mining large amounts of noisy data. Yi et al. (2019) replaced the blocks of the original U-Net with the residual blocks to alleviate the degradation problem and achieve more competitive results with fewer parameters compared to U-Net. Furthermore, (Dixit et al., 2021) organize dilated convolution and residual block as U-Net architecture for *MultiSpectral Images* (MSI). Bakirman et al. provide an experimental analysis of classic CNNs for building extraction in Bakirman et al. (2022). Qiu et al. (2022) fine-tune a pre-trained transformer model with the concept of transfer learning, significantly improving cross-area building extraction performance compared to the original transformer.

Moreover, Huang et al. (2022a) propose a separable factorized residual block and achieve the reduction of the time and space complexity of the model with a slight loss of accuracy. Lin et al. (2019) propose a lightweight network comprised of the basic residual block and the squeeze-and-excitation module. These lightning networks contribute to exploring the practical application of CNNs for building extraction.

3. Challenges and taxonomy

In DL-based building extraction tasks, the challenges are categorized from both model-internal and model-external perspectives. Model-internal challenges refer to those arising from feature representation and learning capacity of the model, mainly concerning the model's effective representation of complex buildings, robustness against environmental interference, and handling of data dependency issues in deep learning. Model-external challenges primarily refer to the upstream data and downstream output of the model, without involving the internal learning mechanisms. They mainly refer to challenges posed by external data and result expression, such as the multi-resolution, multi-sensor, noise, and distortion characteristics of remote sensing images, as well as the requirements for the completeness and practicality of model outputs.

Notably, existing DL research, particularly in computer vision, primarily emphasizes representation and training strategies, while paying less attention to data types and output formats. However, for building extraction tasks, the characteristics of data and output format are crucial for practical applications. Therefore, this survey provides a comprehensive and reasonal summary of the challenges of DL-based building extraction, systematically covering model bottlenecks while addressing the unique demands of the building extraction task.

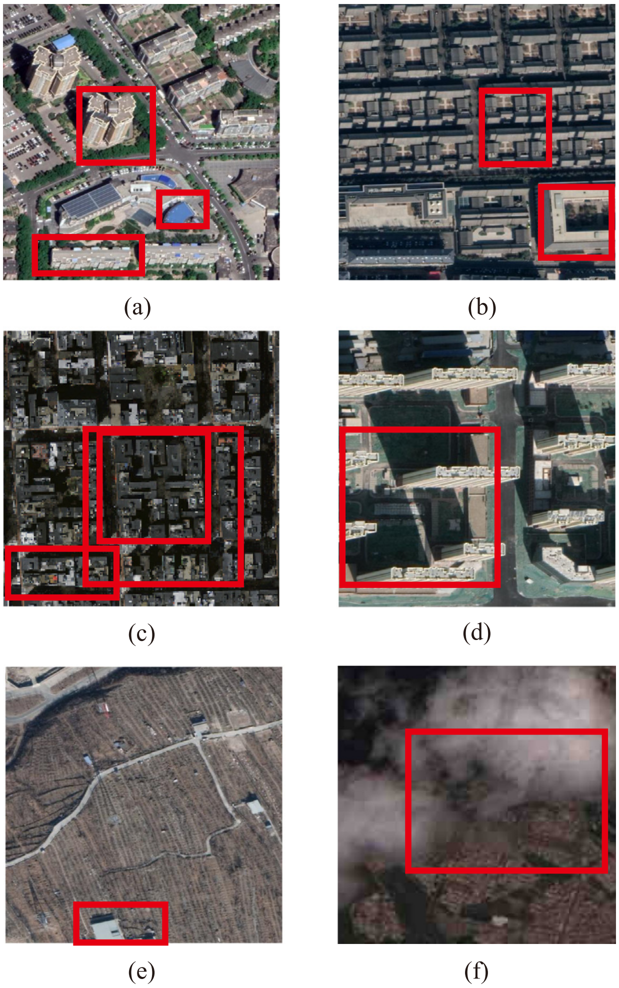

Fig. 6. Some examples that reflect the challenges of building extraction task, from SDLED dataset (Xia et al., 2021), Massachusetts Building dataset (Mnih, 2013) and SEN12MS-CR-TS dataset (Ebel et al., 2022): (a) various geometric features of building, (b) low inter-class variations and large intra-class variations, (c) densely distributed small buildings, (d) shadows and occlusions, (e) the imaged scene with imbalanced samples of foreground and background, (f) cloud cover.

3.1. Model-internal challenges

① Face Visual Complexity of Buildings. One of the core challenges in building extraction is the visual complexity of buildings. Some examples are presented in Fig. 6(a), (b) and (c). Firstly, buildings exhibit a high diversity in geometric shapes, including irregular contours and complex structures, which significantly increases the difficulty for DL models to extract features and achieve accurate recognition. Secondly, high intra-class variance and low inter-class variance further exacerbate the classification challenge. High intra-class variance arises from differences in shape, material, and color due to historical and cultural factors, while non-building elements with significant differences (such as roads, trees, and rivers) are grouped into a single background class. Low inter-class variance refers to the visual similarity between buildings and surroundings, which challenges the model's discriminative capabilities. Lastly, small buildings often suffer from insufficient feature information due to their limited pixel in remote sensing images. And in densely distributed scenes, small buildings are more prone to boundary confusion, leading to decreased segmentation accuracy.

② Under Unfavorable Imaging Conditions. Imaging conditions play a crucial role in building extraction. At first, the presence of shadows and occlusions leads to partial information loss of buildings, making accurate extraction challenging under incomplete observation conditions. Additionally, in remote sensing images of rural, mountainous, or some less urbanized areas, the quantity of buildings is limited. The imaged scene with significant foreground–background imbalance often biases the model towards predicting samples as background, resulting in misclassification. Furthermore, cloud interference is a unique issue in remote sensing images. Cloud cover can significantly degrade image quality, particularly reducing building visibility in optical remote sensing images, thus increasing the difficulty of building detection for the model. Fig. 6(d), (e) and (f) provide an intuitive illustration of these imaging conditions.

③ Data Dependency of DL Paradigm. DL methods are known as data-driven algorithms, and their effective training relies heavily on large-scale, high-quality labeled data. However, remote sensing data processing requires experts with domain knowledge, and constructing datasets for building extraction requires pixel-level or instance-level annotations, which are both time-consuming and costly. The scarcity of annotations significantly limits the performance enhancement of models, particularly the generalization ability.

3.2. Model-external challenges

① Multi-Resolution Remote Sensing Data. The variation in spatial resolution of remote sensing images leads to differences in buildings across scales. Low-resolution images lack detail, often resulting in blurred building edges and information loss. In contrast, ultra-high-resolution images provide rich detail but also introduces a large amount of irrelevant information, increasing computational demands and processing complexity. Due to the multi-temporal characteristics of remote sensing data, buildings and their surroundings may undergo significant changes over time, such as new constructions, demolitions, or seasonal landscape variations, posing challenges for models to maintain consistent extraction performance. Additionally, variations in the spectral and temporal resolution of remote sensing images also pose challenges for building extraction. Low spectral resolution can reduce the model's ability to identify building materials, making it difficult to distinguish buildings from the background. Low temporal resolution may lead to significant environmental changes between observation points, causing shifts in feature distribution and challenging the model's generalization. Conversely, higher spectral and temporal resolution may introduce information redundancy and increase computational burden.

② Multi-Sensor Remote Sensing Data. Remote sensing images can be acquired from various sensors, including optical, SAR, and LiDAR. For single-source input, most existing DL visual models are trained on optical images, and the differences in data patterns across sensors limit the effectiveness of these models on radar data. For multi-source input, overcoming the modality differences and information redundancy among multimodal data is crucial to enhance the effectiveness of data fusion and improve the accuracy of building extraction. The illustration of multi-resolution and multi-sensor data can be found in Fig. 12.

③ Completeness and Practicality of Output. In downstream applications of building extraction, the completeness and practicality of the output are critical. Completeness requires that the extracted building contours remain continuous, avoiding partial omissions, particularly in densely distributed areas. This imposes higher demands on the extraction model. Practicality pertains to the applicability of extraction results in real-world products, such as representing and storing buildings in vector format for GIS applications, thereby requiring effective vector conversion or vector output capabilities from the model.

3.3. Taxonomy of DL-based building extraction

To systematically review DL-based building extraction methods, this study proposes a novel taxonomy. Starting from the methodological motivation and based on the core challenges faced by DL-based building extraction models, existing methods are divided into two main

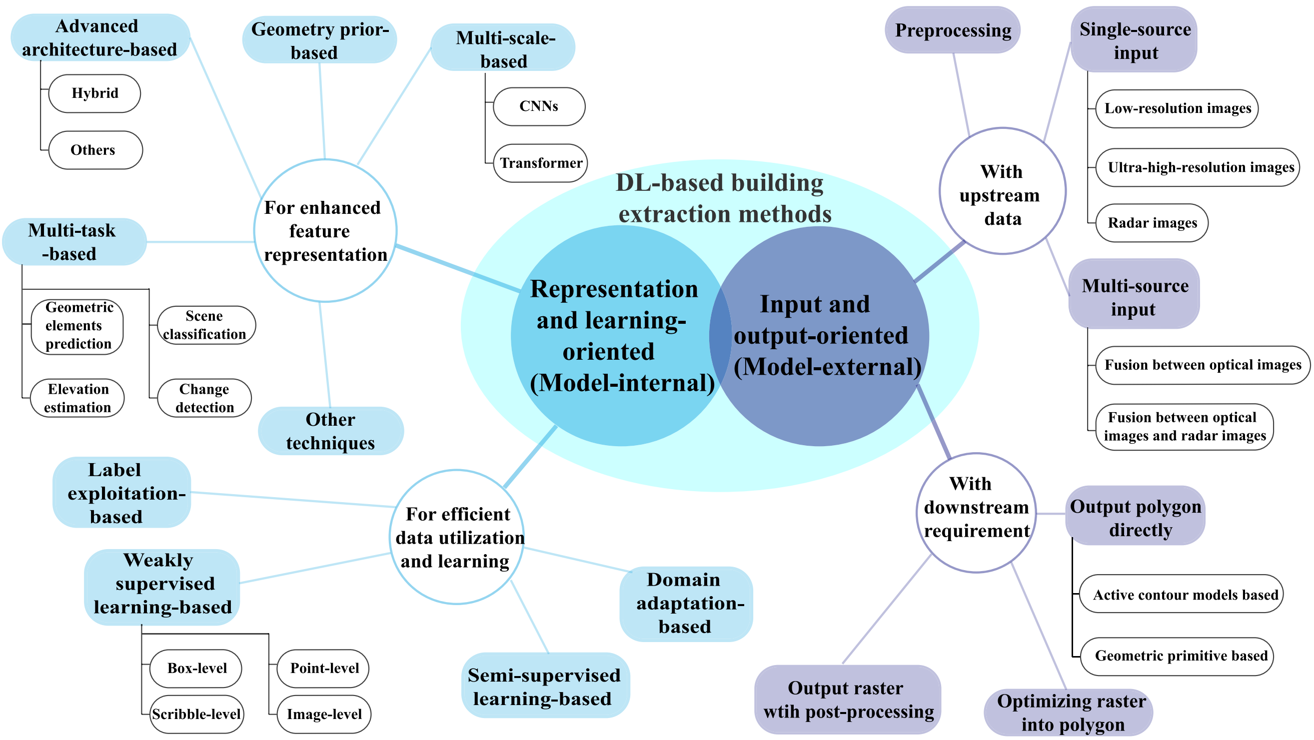

Fig. 7. Taxonomy of the DL-based building extraction methods.

categories: representation and learning-oriented (focusing on model-internal aspects) and input and output-oriented (focusing on model-external aspects), demonstrating the specific pathway through which the technique addresses challenges. The internal and external model perspectives are not isolated or parallel but are interdependent and closely connected. For instance, internal feature representation and learning strategies directly influence the raw output performance of the model, while external data affect the difficulty of feature representation. Thus, under this perspective, the entire process of DL-based building extraction, from data to model to output, is interconnected, forming a complete loop. Moreover, considering the characteristics of building extraction tasks, many studies focus on both data and model, as well as output and model, indicating that these two categories are not entirely independent but have few overlaps.

A detailed illustration of the proposed taxonomy is shown as Fig. 7. For model-internal aspects, researchers tackle challenges through enhanced feature representation and learning strategies to promote the model's feature extraction and generalization ability with limited labeled data. For model-external aspects, researchers primarily address issues through techniques such as super-resolution, multi-source fusion, and directly vectorized output, ensuring the model's effectiveness in real-world scenarios. The proposed taxonomy not only corresponds to the aforementioned internal and external model challenges but also reveals the design rationale and research focus of different methods, clarifying the distinct research paths. In this way, a foundation for systematically evaluating the effectiveness of various methods in addressing specific challenges is provided, thereby contributing to a comprehensive understanding of DL-based building extraction methods and promoting further research.

4. Review of building extraction methods from a representation and learning-oriented perspective

To address the model-internal challenges, researchers focus on optimizing feature representation and learning strategies. Specifically, relevant work includes enhancing representation through multi-scale feature representation, advanced network design, geometric priors, and multi-task learning, as well as employing various learning strategies under data-limited conditions.

4.1. Designs for enhanced feature representation

4.1.1. Multi-scale-based

Multi-scale-based methods primarily address the need for capturing features at various scales in building extraction, particularly when building shapes and sizes vary significantly. These methods enhance the model's understanding of buildings by integrating feature information at multiple resolutions. Specifically, they can be divided into CNN-based methods, which focus on extracting local features, and transformer-based methods, which emphasize modeling long-range dependencies.

① Multi-Scale CNNs. Multi-scale CNNs are widely used for remote sensing image analysis (Liu et al., 2022a) thanks to its ability to comprehensively analyze features. There are two mainstream approaches to implement multi-scale CNNs for building extraction, one is multi-path structure, and the other is well-designed module. An earlier multi-path work is Yuan (2018) which fuse features of different layers to learn more powerful representations for dense prediction. MAP-Net (Zhu et al., 2021), which aims to learn multi-scale features through multi-parallel path architecture, can be seen as a representative work. The features from each path are fused by channel-wise attention. The similar multi-path architecture is widely used by many follow-up work (Zhang et al., 2022a; Wu et al., 2022; Zhao et al., 2023). For well-designed module, dilated convolutions (Ran et al., 2021; Chen et al., 2022b) and attention mechanisms (Liu et al., 2022d; Gong et al., 2023; Dai et al., 2023) are common operations. For instance, Ran et al. (2021) design a pyramid module consisting of multiple atrous convolutions with different rates. They utilize it at the part between the encoder and decoder of U-Net. Liu et al. (2022d) introduce a multi-region attention module to explore long-range relations and use graph-based scale-aware architecture to enhance the understanding of multi-scale features. Moreover, a recent work by Li et al. (2024a) fuses multi-scale features from the perspective of uncertainty prediction, further improving the model performance.

② Multi-Scale Transformer. An earlier study to apply transformer for building extraction is Chen et al. (2021), which designs a multi-branch transformer to learn features from both spatial and channel dimensions. However, the original transformer block lacks the exploration of multi-scale information. To address this limitation, some studies (Chen et al., 2022d; Zhang et al., 2023f) utilize the existing transformer blocks according to multi-scale CNN structure, such as MAP-Net and U-Net, which is a simple yet effective approach. Besides, asymmetric cascade fusion (Chan et al., 2023) is also an with satisfactory performance.

The buildings in remote sensing scenes have various sizes and different distributions that could cause scale variation. Therefore, improving the ability of networks to analyze multi-scale features is a reasonable choice to obtain discriminative representations. Furthermore, in the context of disaster emergency tasks, the presence of damaged buildings with impaired geometric texture features necessitates the model's capability to leverage multi-scale features (Xie et al., 2022). However, it is crucial to note that in the context of remote sensing images, the intricate and fluctuating backgrounds pose challenges to the fusion of features. Inappropriate feature fusion methods have the potential to introduce noise and degrade the overall performance of the model.

4.1.2. Advanced architectures-based

Advanced architectures-based methods aim to meet the need for efficient feature learning in building extraction. These techniques enhance the model's ability to represent diverse building features and improve robustness against imaging environmental interference by introducing hybrid or other novel network architectures.

① Hybrid Architectures. Hybrid architectures effectively combine the efficiency of CNNs in local feature extraction with the powerful capability of transformers in global context understanding, compensating for their respective limitations. Hybridization can be achieved in the following ways: cascading attention and convolution within a block as one branch, followed by feature aggregation with a convolutional branch (Wang et al., 2022a); parallel integration of attention and convolution within a block to improve the original transformer block (Zhang et al., 2023d); and using convolution and transformer as two encoding branches to enable multi-stage feature interaction (Xu et al., 2023; Fu et al., 2024).

② Other Architectures. Besides CNNs, transformer, and hybrid architectures, there are also a few other types of novel architectures used to extract buildings. Zhang et al. (2022b) introduce homogeneous aggregation convolution to pay more attention to sampling points. The capsule network (Sabour et al., 2017; Yu et al., 2022) can be utilized to learn semantically strong and spatially accurate representations.

The introduction of these new architectures facilitates the learning of discriminative features and enhances the robustness of the model. However, they also pose challenges such as increased difficulty in training. Furthermore, the current hardware support for novel architectures like capsule network is not as comprehensive as that for CNNs or transformers. Addressing these issues becomes crucial in order to advance the practical application of these advanced architectures.

4.1.3. Geometry prior-based

Prior knowledge encompasses all additional information pertaining to the problem at hand, beyond the original training data. Buildings have significant geometry disparities, thus the geometry priors play a prominent role as a type of prior knowledge (Song et al., 2024), particularly to boundary and orientation modeling.

There are three specific approaches to leveraging prior knowledge in this context. One approach is to design algorithms that achieve coarse-to-fine refinement on the model's prediction. Common refinement methods are polygon adjustment (Wei et al., 2020), edge points enlargement (Hu et al., 2023b), edge guide (Lin et al., 2023) and iterative subdivision mask (Liu et al., 2023a). Constructing the loss function directly based on the building boundary is another strategy that guides model towards generating more accurate and precise predictions (Jung et al., 2022; Chen et al., 2022e; Lee et al., 2022). It is worth noting that a novel metric called the boundary-oriented intersection over union is proposed in Lee et al. (2022) to evaluate the performance of the predicted boundary. Introducing a specific field is also alternative. For example, Zhou et al. (2022) introduce the direction field to refine the boundary of outputted buildings by the guide of directional relation. Zhang et al. (2023a) design a plug-and-play network head to explicitly predict the direction field. Attraction field representation (Li et al., 2022c) is employed into a model with two sub-network to enrich the boundary information.

Vanilla segmentation methods usually cannot perform well in geometric coverage correctness. As man-made objects, buildings tend to have more prominent and regular geometric features than other ground objects. Thus utilizing the geometry prior to improve the coverage correctness is reasonable and effective. Moreover, in downstream tasks such as 3D reconstruction, the presence of clear boundaries significantly contributes to enhanced modeling of buildings. Nevertheless, the introduction of prior knowledge carries the inherent risk of constraining the generalization ability of the model. Geometry priors are typically designed based on specific assumptions and struggle to generalize to diverse geographical regions or architectural styles. The inherent variability in buildings across different locations can pose more challenges for geometry prior-based methods in unseen environments.

4.1.4. Multi-task-based

Multi-task learning deals with multiple tasks simultaneously, aiming at improving the performance of each task by sharing patterns learned from other tasks. The building footprint extraction task is often combined with geometric elements prediction task (Bischke et al., 2019; Xu et al., 2022a; Guo et al., 2022a; Xu et al., 2022d; Liu et al., 2022b; Xing et al., 2024; Li et al., 2024b), elevation estimation task (Mao et al., 2023b; Lu et al., 2023), scene classification task (Guo et al., 2021a) or change detection task (Hou et al., 2023; Wang et al., 2024b).

① With Geometric Elements Prediction Task. As previously mentioned, geometric priors are a crucial means of optimizing building extraction performance. Consequently, many studies combine geometric element prediction tasks (such as center points and boundaries) with building extraction tasks. For instance, Xu et al. (2022a) introduce a centroid-aware head into the network to predict the geometric center and the mask of the building progressively. Liu et al. (2022b) devise a line segment collaborative segmentation framework that can establish the information interaction between vector-level line extraction and pixel-level segmentation. Xing et al. (2024) propose a sparse sharing multi-task framework based on a dual lottery ticket hypothesis to transfer the learned information from the edge detection task to the building extraction task. Li et al. (2024b) design a decoupling-recoupling module to decouple the multi-scale features into components for boundary prediction task and components for body prediction task.

② With Elevation Estimation Task. Besides building footprints, the elevation of buildings also plays an important role in urban analysis. Recent studies try to unify these two tasks into a single framework. Mao et al. (2023b) utilize the gate mapping mechanism to enhance the interactions between different task features. TCNet (Lu et al., 2023) employs a trident-like decoder for detection, lightweight segmentation, and regression, enabling collaboration between the segmentation and regression results.

③ With Scene Classification Task. Considering the buildings with different spatial distribution, Guo et al. (2021a) exploit the effectiveness of scene prior information by learning the feature for scene classification task and the feature for scene-specific building extraction task.

Table 2
Summary of four types of building extraction methods for enhancing model representation capability.

Method category	Main objective	Basic principle	Advantages	Limitations	Typical scenarios	Representative techniques
Multi-scale	Handle feature representation of buildings at different scales	Extract discriminative information by combining multi-scale feature maps	Improves ability to capture the buildings' diversity of scales and details of	Improper fusion methods may introduce noise, leading to feature degradation	Scenarios where buildings of various scales	Multi-scale CNN, Multi-scale Transformer
Advanced Architectures	Enhance semantic understanding of complex buildings	Integrates multiple mainstream architectures or adopts novel architectures	Enhances model's representation ability to adapt to complex buildings	Challenging making network optimization due to complex architectures	Extraction of buildings with complex shapes or in complicated background scenes	Hybrid architectures, Capsule Networks
Geometric Priors	Utilize geometric structure prior of buildings to improve segmentation results	Incorporate geometric constraints or shape priors into deep learning models	Ensures smooth building boundaries and improves extraction accuracy for regular-shaped buildings	Limited flexibility, applicable mainly to buildings with regular and clear geometric shapes	Extraction tasks focused on regular-shaped buildings	Coarse-to-fine refinement, improved loss function , field guidance
Multi-task Learning	Jointly learn multiple building-related tasks to enhance feature generalization	Share feature representations and optimize related tasks collaboratively	Leverages synergy between building extraction and related tasks, more generalized performance	Task conflicts may complicate model optimization	Scenarios with available auxiliary label of needs of information beyond building footprints	Multi-model collaboration, shared encoder architecture

④ With Change Detection Task. Given that both building extraction task and change detection task rely on the discriminative building representation, Hou et al. (2023) constructs a multi-task framework for performing these two tasks based on object-wise prototypes, improving the accuracy of building extraction as well as exploring the single-temporal supervised change detection models. Wang et al. (2024b) employ the foundation model with task prompts to unify the representation for building extraction and change detection, showing remarkable generalization.

Through learning sharing patterns of complementary tasks, the performance of building extraction in diverse scenarios can be improved. Moreover, the practical application requirements can also be met. Specifically, for urban planning, obtaining not only accurate building footprints but also information about the scene categories of building clusters holds significant value. This particular requirement involves the integration of both the building footprint extraction task and the scene classification task. However, multi-task learning is limited to the potential interference between tasks. For instance, footprint extraction primarily emphasizes the delineation of individual building boundaries, whereas scene classification targets the analysis of building distribution and their potential relationship with the surrounding background. Effectively distinguishing between task-independent features and task-shared features still presents a significant challenge.

4.1.5. Other techniques

The four types of methods mentioned above enhance the model's feature representation capability through architecture design or auxiliary information. A comparative summary of these methods, including but not limit to their advantages, limitations, and typical scenarios, is presented in Table 2. In addition to the techniques mentioned above, here are some very enlightening technologies to learn more discriminative features for building extraction, supplemented as follows, including adversarial learning, feature decoupling, model ensemble, and utilizing multi-view information.

For adversarial learning, Ding et al. (2022) introduce the adversarial learning strategy to enforce the network to concentrate on the shape information. Aiming to suppress a lot of background noise, Guo et al. (2023) use the feature decoupling method to decompose semantic and edge representations from low-level features. Model ensemble technology is utilized by Yang et al. (2023a) to address the issue of imbalanced samples of foreground and background. With multi-view images available, Workman et al. (2022) present a geospatial attention module to capture the geospatial relation between an overhead image and nearby ground-level panorama. More fine-grained information for building extraction and analysis are exploited by this attention module. Qi et al. (2023) obtain novel view segmentation of building from multi-view images and limited annotations by combining 2D CNNs features with the neural field 3D features.

4.2. Strategies for efficient data utilization and learning

There are four kinds of approaches to alleviate data dependency in DL-based models. An overall comparison of these methods is illustrated in Fig. 8 to present their differences more intuitively.

4.2.1. Label exploitation-based

To deal with insufficient training data, exploiting existing labels is a spontaneous approach. Data augmentation is widely applied to improve generalization ability. For instance, Illarionova et al. (2021) devise object-level augmentation which integrated building segmentation mask and background patch to synthesize new samples for network training. Wangiyana et al. (2022) report the influence of several data augmentation methods on the performance of building extraction with *Synthetic Aperture Radar* (SAR) data, which can provide potential guidelines for future research. Luo et al. (2023) combines traditional data augmentation methods with style transfer methods to expand data at the level of domain generalization. In addition, learning from noisy labels is also a feasible way. Note that *Open Street Maps* (OSM) (Web, 2023b) can provide amounts of noisy building information that may lead to confusion, Kang et al. (2022a) propose a joint loss function for training the network with missing annotations.

These methods effectively extend the volume of available data. However, they are constrained by the adequacy of the sample's diversity and its susceptibility to noise interference, thereby presenting challenges in achieving satisfactory performance on real-world data.

4.2.2. Weakly supervised learning-based

One of the reasons that semantic segmentation data annotation is expensive and time-consuming is that it needs to be labeled at pixel level. Intuitively, if the network can be trained with coarse-grained labels, the burden of annotation will be eased effectively. Hence the researchers explore the performance of building extraction using weak labels (image-level, scribble-level, point-level and box-level), known as

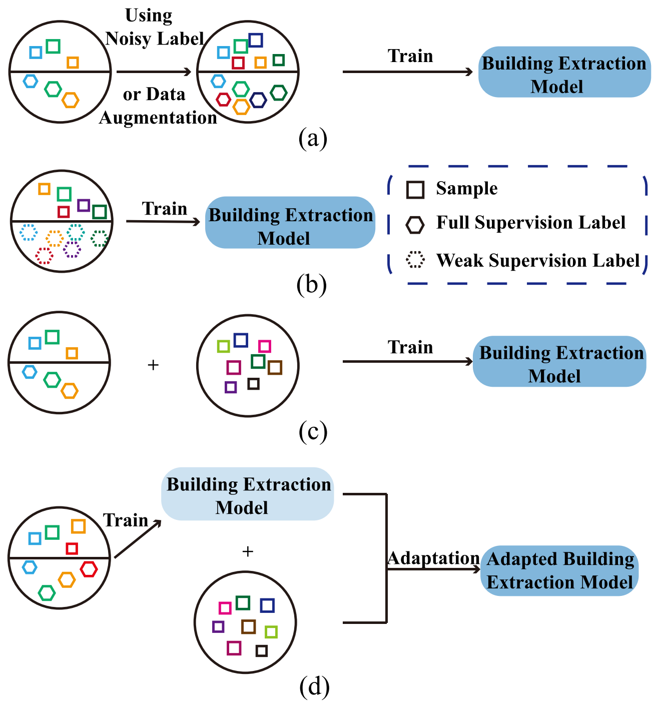

Fig. 8. Illustration of four typical methods for processing limited dataset: (a) label exploitation-based methods, (b) weakly supervised learning-based methods, (c) semi-supervised learning-based methods, (d) domain adaptation-based methods.

weakly supervised learning. An intuitive illustration of these four types of weak supervision labels is shown in Fig. 9.

① Box-Level Label. By drawing bounding boxes around buildings in the image, this kind of label provides location information while reducing labeling complexity, retaining basic spatial details. For specific study, Zheng et al. (2023) present a multi-scale feature retrieval module to suppress the noise from the bounding box background.

② Scribble-Level Label. Scribble-level label marks building regions with simple lines or scribbles, lowering the demand for precise annotations. Under this setting, Chen et al. (2022a) propose a scribble extension module and a loss function based on edge structure to reconstruct buildings from scribbles.

③ Point-Level Label. This type of annotation involves marking only the center or key parts of buildings with points, providing minimal annotation information and greatly reducing the labeling burden. Correspondingly, it places higher demands on training strategies. For point-level label, Lee et al. (2021a) consider large building objectives and small building objectives separately, and employ simple points to label small objects. This method of point labeling cannot only reduce the labeling burden but also improve the extraction accuracy of small objects.

④ Image-Level Label. This label assigns category to entire images based on the presence or absence of buildings. Although the annotation is coarse, training models with large-scale data has also proven effective by leveraging the diversity of images. For example, Fang et al. (2022) refine pseudo-mask through fusing optimized class activation map with semantic affinities. Yan et al. (2022) simply merge multi-scale *class activation maps* (CAMs) from multiple generation modules and polish it with super-pixel. Wang et al. (2023b) aggregate multi-scale CAMs via self-attention and propose a scale-invariant optimization module to enhance the completeness of CAMs.

The flexibility of weakly supervised learning in accommodating various annotation formats is evident. Nevertheless, it is important to acknowledge that remote sensing images often exhibit intricate scene compositions, rendering the limited supervision provided by weakly

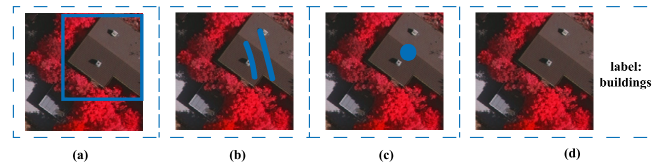

Fig. 9. Illustration of four types of weak supervision labels: (a) box-level label, (b) scribble-level label, (c) point-level label, (d) image-level label.

supervised methods may be insufficient for effectively discerning such complexity.

4.2.3. Semi-supervised learning-based

Semi-supervised learning is an approach that leverages a small amount of labeled data and a large amount of unlabeled data to train a model. There are many works apply semi-supervised learning to building extraction, each with its unique approach. Xia et al. (2021) generate pseudo labels with the guide of a segmentation model pre-trained on labeled datasets and then fine-tune the model with both manual labels and pseudo labels. Kang et al. (2021) focus on the pixel-level latent representation contrast between buildings and non-buildings. Guo et al. (2021b) apply semi-supervised learning to bi-temporal images, updating existing building footprints with a few manual labeling. Li et al. (2022e) employ a consistency learning strategy and image-specific perturbation to train a shared encoder, a main decoder, and an auxiliary decoder.

Compared with visual data in other fields, remote sensing is characterized by abundant accessible data. Therefore semi-supervised learning is more suitable for building extraction under the condition of limited annotation data. Considering the challenges of building extraction, caution should be exercised when utilizing unlabeled data to prevent the introduction of excessive negative samples that may exacerbate the imbalance issue between foreground and background classes.

4.2.4. Domain adaptation-based

Although the above methods reduce the annotation burden, they still require man-made labels. Note that there are several publicly available building extraction datasets from various scenes (shown in Section 6), domain adaptation is also a suitable choice to reduce the labor of annotation. *Unsupervised domain adaptation* (UDA) is to adapt a model trained on source domain (with labels) to target domain (different distribution but same task, without labels).

Some representative studies of UDA for building extraction are following. Self-mutating network (Lee et al., 2021b) is proposed to adjust the parameters in response to the domain of an input aerial image. Na et al. (2021) introduce both domain transfer methods and adversarial attack methods to convert source domain image to target domain style. Chen et al. (2023a) utilize a normalization-based image style transfer strategy and a memory-contrastive module to obtain domain-invariant features. Subsequent work (Chen et al., 2024b) further propose decoupling style and semantic features to minimize interference from style information and enhance the effectiveness of memory mechanisms in cross-domain building extraction tasks. Wang et al. (2023c) present a framework that provides more positive support for adaptation by selecting, purifying, and exchanging domain information to achieve better adaptation.

Domain adaptation is a popular technique to cope with changes in the target scenario. However, it is crucial to acknowledge that the process of domain adaptation may lead to the inadvertent loss of essential discrimination features, consequently resulting in a degradation of performance. Thus, in instances where the dissimilarity between scenes is significantly different, selectively labeling some data may be a judicious choice.

Table 3
Summary of four types of building extraction methods for efficient data utilization.

Method category	Main objective	Technical principle	Advantages	Limitations	Typical application scenarios	Representative technologies
Label Exploitation	Maximize the efficiency of utilizing existing labeled data	Extract information from existing labels through innovative label utilization methods	High label utilization efficiency without adjusting the model, low model complexity	Requires careful label utilization strategy design, risk of overfitting or learning noise	Data with diversity, datasets with noisy labels	Data augmentation, noisy label learning
Weakly Supervised Learning	Train models using non-pixel-level weak annotations	Guide the model to infer detailed building objects through carefully designed training strategies	Significantly reduces annotation workload, some special forms of annotation may benefit specific scenarios (e.g., point for small buildings)	Extraction results may have significant deviations, complex model training	Large-scale rough building extraction	Box-level, scribble-level, point-level, image label-level annotations
Semi-supervised Learning	Improve extraction performance by combining a few labeled data and abundant unlabeled data	Utilize the implicit information provided by unlabeled data under the assumption that the distribution between labeled and unlabeled data is consistent,	Reduces dependence on labeled data, leverages learnable latent distribution in unlabeled data to improve generalization	Unstable training in early stages, noise introduced by the differences between labeled and unlabeled data	Scarcity of labeled data but availability of a large amount of similarly distributed unlabeled data	Pseudo-labeling (self-training), consistency learning
Domain Adaptation	Enable the model to generalize well in the target domain without annotations	Align features or match distributions between the source and target domains to learn domain-invariant features	Improves adaptation between different remote sensing datasets or regions without additional annotation	Relies on source domain annotation quality, performance degrades when domain discrepancy is large	Cross-region, cross-sensor building extraction tasks	Feature alignment, adversarial learning

The aforementioned four types of methods provide valuable insights for training models with limited data. A summary of their advantages, limitations, and applicable scenarios is presented in Table 3. However, the reduction in data dependency usually comes at the expense of a potential decrease in accuracy compared to models trained with ample labeled data. In an effort to alleviate the labeling burden without compromising accuracy as possible, foundation vision models have emerged as a promising solution for the future, which will be discussed in Section 7.

5. Summary of building extraction methods from an input and output-oriented perspective

To address model-external challenges, researchers focus on ensuring that input data with appropriate resolution and rich information patterns, as well as guaranteeing the accuracy and practicality of the output. Specifically, related work includes super-resolution, ultra-high-resolution image processing, multi-source data fusion, and techniques such as polygonization and vector output. In this section, the methods are organized from the perspectives of inputs (upstream data) and outputs (downstream requirements). First, the related models are grouped according to whether the input data is multi-source or not. Then, the methods are summarized according to which format their outputs is in.

5.1. Upstream data-oriented methods

5.1.1. Preprocessing for raw data

In building extraction tasks, raw data preprocessing is a key step to enhance the performance of visual models. Given that this survey focuses on DL methods, preprocessing techniques are divided into two main categories: traditional preprocessing for remote sensing images and data augmentation commonly applied in DL. The former includes techniques such as pan-sharpening and image denoising, while the latter emphasizes augmentation methods like scaling and flipping.

The traditional preprocessing techniques for remote sensing images focus on enhancing the usability of raw images, ensuring that the model receives high-quality images for extracting building features. For instance, in Huang et al. (2022b), pan-sharpening is used to fuse multispectral images with panchromatic images, improving spatial resolution and the clarity of building details. In Li et al. (2023a), an adaptive method is employed for SAR image denoising to eliminate inherent speckle noise, thereby improving image quality and enhancing the distinguishability of building features. The data augmentation methods integrated with DL models aim to enhance model generalization and robustness. For example, in Wang et al. (2022a), data augmentation strategies such as horizontal and vertical flipping (random at training phase, no random at test phase) are employed to simulate variations in buildings under different perspectives and scales, extending the model's adaptability to diverse inputs. In Wei et al. (2023), augmentation methods including multi-scale scaling, cropping, flipping and color jittering are introduced to enrich training samples, improving the generalization of the model. The combination of traditional preprocessing techniques and data augmentation enhances the quality of input data, supporting accurate extraction of building features by the model.

5.1.2. Single-source input

Methods based on single-source data mainly use optical, *Light Detection And Ranging* (LiDAR) images (this survey focuses on 2D representation), or SAR images. A brief analysis of the advantages and disadvantages of these three types of data on the building extraction task is summarized, as shown in Table 4.

This review further divides single-source input methods into three parts: super-resolution for low-resolution images, methods for ultra-high-resolution images, and methods for LiDAR/SAR images. The first two parts primarily focus on spatial resolution of the image, with related research mainly based on optical data, thus warranting a separate review for each. It is also worth noting that remote sensing images exhibit multi-resolution characteristics in spatial, spectral, and temporal dimensions. Images with high spectral resolution usually have low spatial resolution. A common approach is to perform pan-sharpening

Table 4
The comparison of three types of single-source data used for building extraction.

Data type	Data characteristics	Advantages	Limitations
Optical	Passive imaging in various bands, reflecting the surface reflectance properties of objects	Provides rich spectral, color, and texture details that intuitively represent building features	Highly affected by environmental factors such as illumination, weather, and cloud cover
LiDAR	Active sensor based on lasers, directly providing height information of the buildings	Accurately capture elevation differences between buildings and surrounding objects, suitable for high-precision modeling	Large data volume, limited by the sampling density and angle of the laser sensor
SAR	Active microwave imaging, capable of penetrating clouds and collecting data in all weather conditions	Unaffected by illumination and weather, capable of capturing microwave features of buildings	Data usually contains a large amount of speckle noise, geometric information of buildings is not intuitive

during data preprocessing or use the dual-branch CNN-based multi-source fusion methods introduced later. As for the studies related to the multi-temporal characteristics, they primarily involve the previously mentioned multi-task learning methods (change detection Hou et al., 2023; Wang et al., 2024b), and are not repeated here. The related research has already been discussed in other sections. Therefore, it will not be repeated here from the perspectives of spectral and temporal resolution. Additionally, since most works discussed in Sections 3 and 4 involve high-resolution or medium-resolution optical images, this subsection does not elaborate further on those but instead focuses on supplementing studies related to low-resolution and ultra-high-resolution optical images. For the third part, current DL-based visual techniques are predominantly designed for optical data, which differ significantly from radar images, resulting in relatively fewer studies applying DL-based visual techniques to radar data. Therefore, radar is reviewed separately as a distinct part.

① Super-Resolution for Low-Resolution Optical Images. In practice, the spatial resolution of remote sensing images significantly impacts the effectiveness of existing segmentation models (Liu and Tang, 2023), particularly when there is a substantial difference between the resolution of training and testing data. Additionally, in low-resolution optical images, building details and edges are often blurred. To this end, super-resolution techniques can provide richer information for subsequent extraction of building features by enhancing spatial resolution. One type of studies integrates super-resolution and segmentation models into a multi-step framework. For example, Guo et al. (2019) design a framework to adapt super-resolved low-resolution images into a super-resolution space, alleviating the resolution difference between training and testing images. Chen et al. (2023c) propose a pipeline organizing scene classification, super-resolution, building instance segmentation, and image mosaicking in a cascade fashion. Other studies combines super-resolution and segmentation as different modules within a single network. Specifically, Zhang et al. (2021a) adopt a two-stage approach, while Xu et al. (2021) achieve this in an end-to-end manner. Moreover, Wang et al. (2024a) employ a teacher-student paradigm to extract priors from super-resolved images and apply them to low-resolution images. Besides directly aiding fine-grained feature extraction by the model, super-resolution also facilitates the monitoring of urbanization over a wider time span. It enhances the resolution of historical images, which are typically lower in resolution, allowing for more accurate and comprehensive analysis of long-term urban development.

② Methods for Ultra-High-Resolution Optical Images. Ultra-high-resolution images contain abundant details and complex textures, but their large size often exceeds memory capacity for direct processing, while cropping may disrupt the integrity of building features. Thus, methods are required to balance computational constraints with the completeness of building features. Chen et al. (2019) design a dual-branch collaborative network, where the global branch takes the downsampled entire image as input, and the local branch uses cropped patches as input. The features of these two branches interact in a bidirectional manner. Li et al. (2021b) propose a local segmentation strategy utilizing locality-aware contextual correlation to handle regions where local patches and their contextual semantics exhibit significant variation. ISDNet (Guo et al., 2022b) introduced by Guo et al. processes the entire image directly, which fuses shallow features of the whole image and deep features of the downsampled image via a relation-aware feature fusion module. Ji et al. (2023) use the mask of the downsampled image to guide patch grouping and employ a lightweight transformer to maintain contextual dependencies while further reducing computational burden.

③ Methods for LiDAR and SAR Images. LiDAR and SAR images, due to their unique imaging mechanisms, exhibit visual characteristics distinct from optical images. Related researches primarily focus on leveraging the specific properties of radar data. For LiDAR data, Shin et al. (2022) focus on the additional characteristics of LiDAR (like multiple returns) and achieve effective building extraction from point clouds. Soliman et al. (2022) train a U-Net utilizing *Digital Elevation Model* (DEM) under the supervision of noisy labels. Vats et al. (2024) propose a terrain-aware self-supervised learning method based on LiDAR data, improving the model's extraction performance under limited annotations. For SAR data, Sun et al. (2022a) predict build-up areas by utilizing multilevel SAR features with the auxiliary of inaccurate GIS data. Chen et al. (2022c) design a complex-valued convolutional network with a multi-feature fusion strategy to extract buildings from interferometric SAR images. Zhang et al. (2024a) utilize a network with a large receptive field to address the low signal-to-noise ratio in SAR images and introduce an auxiliary branch to reduce classification errors.

5.1.3. *Multi-source inputs*

Generally, the information provided by different sources is highly complementary (Audebert et al., 2018; Zhang et al., 2024b). This characteristic makes it possible to analyze building objects at a comprehensive level. Next, recent studies are reviewed from the perspective of data combination. One is the fusion between optical images, and the other is the fusion between the optical images and the radar images. A brief summary of these two types of methods is presented in the Table 5.

① Fusion between Optical Images. Due to the limitation of the imaging mechanism, the *PANchromatic* image (PAN) are in higher

Table 5
The comparison of two types of multi-source data fusion used for building extraction.

Method category	Fusion objective	Advantages	Limitations
Optical-Optical Fusion	Enhance spatial and spectral features by utilizing information from different spectral bands	Retains high-resolution and multispectral information, enhancing spatial and spectral details	Dependent on lighting conditions, easily affected by cloud cover and weather
Optical-Radar Fusion	Combine information from heterogeneous imaging	Optical and radar data are complementary, improving adaptability to complex scenes	Significant differences between data sources, making fusion challenging

spatial resolution, while the *HyperSpectral Image* (HSI) and MSI are in higher spectral resolution. The comprehensive use of them can obtain rich correlationship features between spatial information and spectral information, which provides a basis for more accurate building extraction. Considering the resolution difference between the two types of images, the main practice is to extract features by parallel CNNs after preprocessing. Huang et al. (2016) employ a pan-sharpening method to integrate spatial information of PAN and spectral information of MSI. The pan-sharpened MSIs are then split into two band combinations where two parallel CNNs is applied respectively. Chen et al. (2020) process PAN and MSI into superpixels at first and then extract multi-modal features with a multispectral CNN and a panchromatic CNN.

② Fusion between Optical Images and Radar Images. As acquired by the active imaging technique, the radar image is not restricted by light, weather, and other factors. Meanwhile, the radar image provides more geometry information and is not impaired by shallows. However, the radar image is lack of texture features. Thus integrating optical images and radar images is also a powerful approach to boost the performance of building extraction.

Many work focus on the comprehensive utilization of multi-modal features. For example, Zhang et al. (2020) design HAFNet consisting of three parallel branches for RGB data, LiDAR data, and cross-modal data to exploit the individual and cross-modal features of buildings. Hosseinpour et al. (2022) propose CMGFNet to learn RGB and LiDAR information with two separate encoders. In addition, they employ two different strategies for cross-modal feature fusion and multi-level feature fusion. Kang et al. (2022b) propose the DisOptNet to distill the knowledge of the network pre-trained with optical images. They generate segmentation results with only SAR images for the testing phase. Zheng et al. (2021b) design a prototype network to learn meta-sensor representation for RGB and SAR images. Li et al. (2023a) integrate the deep features of SAR images, the phase, and optical images by a progressive fusion learning approach. To fuse multi-modal features more effectively, Yuan and Mohd Shafri (2022) present a local perceptron that can achieve mutual compensation of spatial and spectral features. Luo et al. (2024) employ transformers as encoder and fuse cross-modal features via self-attention mechanism. Chen et al. (2024a) propose a refinement module to refine fused cross-modal features at multi-levels and introduce a residual upsampling module for feature recovery.

A few work focus on more flexible utilization of multi-modal data. Xie et al. (2023) present a co-learning framework which learns modal features with a soft connection so that paired data is unnecessary during the test phase. Considering that multi-source data may not be obtained simultaneously, Xia et al. (2022) propose two types of pipelines for paired and unpaired RGB-SAR datasets to explore the possibility of using different modal data in the training and inference stages.

The all types of fusion methods discussed above have demonstrated remarkable advancements, resulting in enhanced accuracy even in challenging scenarios characterized by complex ground objects composition and harsh imaging conditions. Nonetheless, it is worth noting that the majority of these methods primarily emphasize the effective fusion of multi-source features, often ignore the limitations associated with the practical application of multi-source data (discussed in Section 7). To foster the practical applicability of these methods, it becomes imperative to devise strategies that effectively mitigate these constraints.

5.1.4. Single-source input vs. multi-source input

More types of data provide richer information, but this does not mean that the methods using multi-source data as input is always better than the methods using single-source data.

Indeed, the complementary characteristics of the multi-source data provide a more diverse and comprehensive pattern of objects, which is beneficial to improve the accuracy and robustness of the model. And from the perspective of downstream application, incorporating novel types of data holds the potential to significant enhancement, such as night light data for the realm of population estimation. Nevertheless, the use of multi-source data also inevitably brings some difficulties. Specifically, more data requires more operations, which increases the computational burden. The features of multi-source data are not only complementary, but may also diverge from each other. This potential conflict will damage the optimization of the model. In addition, multi-source data need to be preprocessed where errors often occur (Uss et al., 2016).

Accordingly, although the methods based on single-source data are limited by the non-plentiful feature pattern, these methods still have some advantages. On the one hand, there are more open access single-source datasets available for model training. On the other hand, these methods have less overall computational burden and are less restricted by the network scale when introducing new modules.

5.2. Downstream output-oriented methods

5.2.1. Post-processing for raster prediction

Since the well-established deep semantic segmentation models employ pixel-to-pixel strategies, the methods mentioned in Sections 2 and 4 are directly output in raster formats like Fig. 10(a). Here, this survey does not be repeated this kind of methods but briefly summarizes raster-based post-processing techniques applied with deep visual models.

Raster post-processing is a crucial step connecting the model's raster prediction to the high-quality results, aiming to enhance the completeness and accuracy of initial segmentation outputs without additional training. The common post-processing techniques for outputs of DL models include threshold adjustment, area-based *non-maximum suppression* (NMS), and regularization. At first, deep segmentation models output a probability map indicating the likelihood of each pixel belonging to a building. A threshold is required to convert this probability map into extraction results. Thus, extraction accuracy can be optimized by adjusting the threshold. Li et al. (2019) determine the optimal threshold based on different cities and prior knowledge of building sizes in the validation set, achieving satisfactory results. Moreover, raw remote sensing images are typically processed by splitting them into

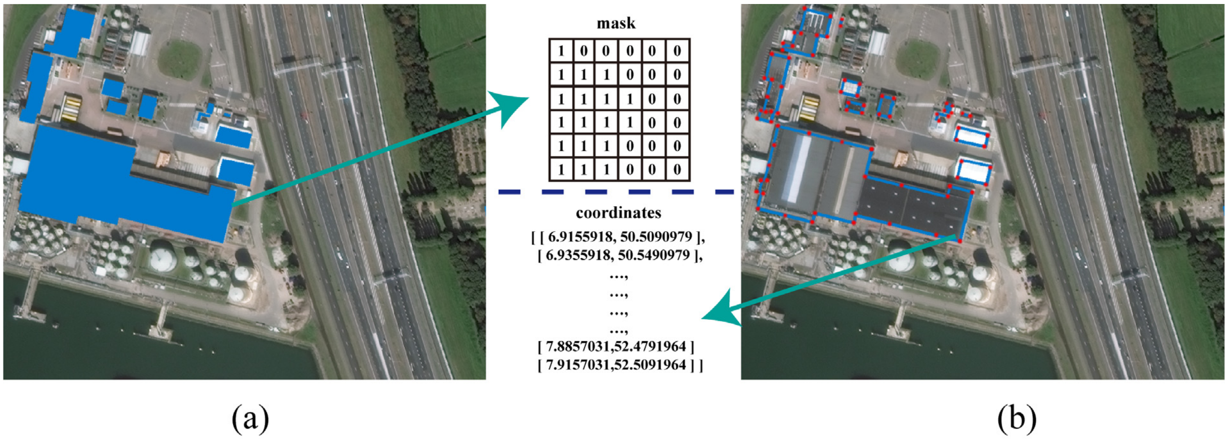

Fig. 10. Different output format: (a) raster output, (b) polygon output.

smaller patches using a sliding window. Segmentation results tend to have errors at the edges of these patches. To this end, one feasible approach is to maintain a certain overlap between adjacent patches and apply NMS algorithm to these overlaps when merging extraction results. Considering the characteristics of building extraction tasks, an area-based NMS (Chen et al., 2023c) is usually employed, where masks in overlapping areas are sorted by their size (instead of scores as in traditional NMS) and then filtered intersection-over-union values. Additionally, to ensure that the extraction results conform to the actual shape of buildings, regularization techniques can be applied to address issues like geometric artifacts. For instance, Zhu et al. (2024b) employ an area compression algorithm to correct the deformation of building masks.

5.2.2. Optimizing raster prediction into polygonal prediction

As illustrated in Fig. 10(b), a building instance can be represented appropriately by a polygon consisting of a set of vertices and edges. This representation can be stored in vector format that commonly used by GIS application. The raster prediction usually needs to be converted to polygon format in practice. Generally, during the conversion of the raster mask to polygon format, the bias generated by the raster output model will be passed down, resulting in the final irregular and deviated polygons. To this end, researchers have proposed many DL-based approaches to produce more accurate polygons.

One immediate idea is to optimize the boundaries of the raster results. Zorzi et al. (2021) refine the raster prediction through the adversarial strategy, so that the simple polygonization algorithm can achieve good performance. More generally, the previously introduced geometry prior-based methods are also suitable choices for outputting better polygons. Another series of work focuses more on how to optimize polygonization process. For example, Li et al. (2020) utilize local merging and splitting operations to partition polygon, and Girard et al. (2021) introduce frame field to optimize the contour alignment. Moreover, multi-task learning framework is utilized to provide more information, like edge orientation (Li et al., 2021c, 2023b) or attraction field (Xu et al., 2022c), for polygon refinement.

These methods yield predictions with more details, offering improved support for applications like GIS maintenance and 3D reconstruction. Nevertheless, these techniques increase network complexity, and the non-end-to-end operations within some of them poses challenges in ensuring the generation of optimal polygon results.

5.2.3. Output polygonal prediction directly

Although there are many means to obtain the polygon format building mask from raster output, it is still challenging to compensate for the loss of detail features. Thus some work have focus on generating the building polygons in an end-to-end manner. These works can be

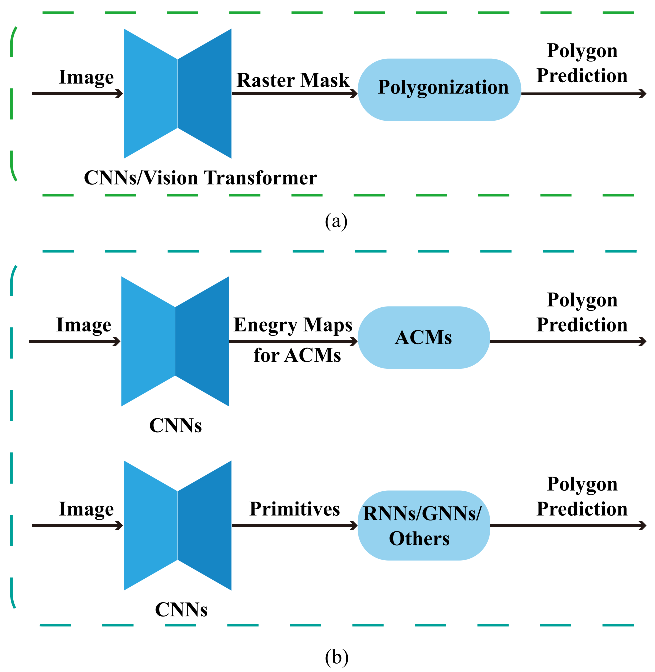

Fig. 11. Different approaches to obtain polygon prediction: (a) raster to polygon, (b) output polygon directly.

summarized as two categories: *Active Contour Models*-based (ACMs-based) and geometric primitive-based. A clear comparison of different methods to obtain polygon results, e.g., conversion based on raster and direct prediction, is illustrated in Fig. 11.

① ACMs-Based. Kass et al. propose a popular image segmentation approach named *snake* (Kass et al., 1988), known as the first ACMs. The key idea of ACMs is to encourage contour movement to minimize the energy function which formulates intuitive geometry priors and constraints.

With the rise of DL, researchers attempt to combine CNNs and ACMs to obtain more expressive and flexible building extraction models. The fundamental concept revolves around leveraging CNNs to generate energy function maps, which are subsequently utilized by ACMs to predict polygons. Specifically, Marcos et al. (2018) devise a deep structured active contour model to predict the parameters of the energy function by CNNs. Cheng et al. (2019) further extend this design to polar coordinates to avoid contour self-intersection. Hatamizadeh et al. (2020) utilize CNNs not only to produce parameters for energy function

Table 6
The comparison of three types of methods completeness and practicality of building extraction results.

Method category	Processing workflow	Advantages	Limitations
Raster Output with Post-processing	First predict rasterized building areas, then apply post-processing (e.g., polygonization) to obtain vector output	Easy integration with existing raster-based methods; post-processing can correct boundary errors to some extent	Post-processing depends on the quality of the initial raster prediction, not end-to-end
Optimizing Raster into Polygon	Extract polygons from raster predictions, and use optimization algorithms or independent module models to improve polygon quality and generate vector output	Effectively corrects boundary errors in raster output, improving geometric consistency of vector results and enhancing accuracy	Increases processing complexity, not fully end-to-end
Direct Polygonal Output	Directly predicts building polygon boundaries from input imagery, avoiding the rasterization step	Produces vector output directly, offering an end-to-end solution and avoiding boundary biases introduced by rasterization	Limited by training data, may struggle with complex environments or buildings with intricate geometric shapes

but also to initialize the ACMs. Inspired by physical string vibration theory, Xu et al. (2022b) design a novel ACMs called CVNet which evolve contour through mechanical equations of motion.

Such methods typically rely on iterative processes to learn optimal boundaries. But the final results are heavily influenced by the polygon initialization. While they have shown promising results in tests conducted on small images featuring prominent buildings in the center, their performance is still constrained when applied to larger images or complex scenes.

② Geometric Primitive-Based. Geometric primitives usually refer to edges, corners, and regions. The building polygon can be obtained by inferencing based on these geometric primitives. The direct pipeline is to use CNNs to detect primitives at first and then to reason polygons out with *Recurrent Neural Networks* (RNNs) (Cho et al., 2014) or *Graph Neural Networks* (GNNs) (Welling and Kipf, 2016). In terms of RNNs, Zhao et al. (2021) employ stacked convolutional gated recurrent units to reduce the computational burden, while Huang et al. (2022c) delineate outlines of building with predicted edge attention map. For GNNs, Zhao et al. (2022) complete the relation reasoning module based on the line segment features and the semantic features. Zorzi et al. (2022) use attentional GNNs to capture the relationships of vertex with position information and visual features. Yang et al. (2023b) design a class-agnostic framework that robustly formulates both polygon-shape and line-shape targets by dynamically generating the ground truth of adjacency graph label.

Besides the paradigm above, there are some other well-designed strategies to predict building polygons. Nauata and Furukawa (2020) formulate detected primitives and their relationship as an integer programming problem. Zhang et al. (2021b) propose an explore-and-classify framework based on the heuristic search strategy. Zhu et al. (2022) adaptively select the vertex permutation most similar to the individual building. PolyBuilding (Hu et al., 2023a) performs building instance detection and polygon generation in parallel by using transformer-based model. BISVP (Zhang et al., 2023b) predicts serialized vertices of buildings directly in a multi-scale and bidirectional manner. Hu et al. (2024) propose a shape-constraining module that fuses Fourier descriptors based on mask features with vertex and boundary features integrated through RNN. Zhu et al. (2024a) introduce a iterative polygon deformation algorithm that predicts point offsets on the initial contour and updates the point set through set matching to correct missing vertex.

Compared to ACMs, this kind of methods demonstrate greater effectiveness in handling large images with multiple buildings. However, a significant challenge for model optimization arises due to the relatively small proportion of geometric primitives within the entire sample set.

5.2.4. Raster output vs. polygon output

Although the three methods mentioned above yield different output formats, they all aim to optimize the completeness and practicality of extraction results. A brief summary of their advantages and limitations is presented in Table 6. Below, a comparative discussion of the raster output and polygon output is provided.

The raster output achieves high accuracy at the pixel level and also covers the general area of the building well. Raster output corresponds to the definition of semantic segmentation. Compared with polygon output, a large number of sophisticated raster output models are available for users to choose from. And since there are no prior geometric constraints, raster output methods can learn representations of buildings with complex shapes (*e.g.*, hollow interiors). However, it increases the difficulty of learning the boundary features of buildings as well. In addition, raster output needs to be polygonized before it is accepted by downstream geo-information applications. This process may also introduce bias.

On the contrary, polygon output methods can be directly applied to the GIS, and its extraction results also have clearer boundaries and sharper edges. But this also limits its ability to handle buildings with complex shapes. At the same time, the modeling process of the polygon output method is sensitive to the initialization of contours or primitives.

Another important topic for discussion is the relationship between different output formats and the prediction of instance-level building footprints. In the case of polygonal outputs, it can be assumed that each polygon corresponds to an individual building instance. In an ideal scenario, this format provides instance-level information accurately. Only considering raster output formats, they do not directly provide instance-level information. The raster format represents buildings as binary pixels, offering only the overall distribution information without distinguishing different building instances. However, if a raster prediction is obtained using instance segmentation methods (Xu et al., 2022a; Hu et al., 2023b), the model can distinguish different building instances. Furthermore, for more accurate raster prediction, a connected region can be approximated as a building instance, particularly when there is clear separation between buildings.

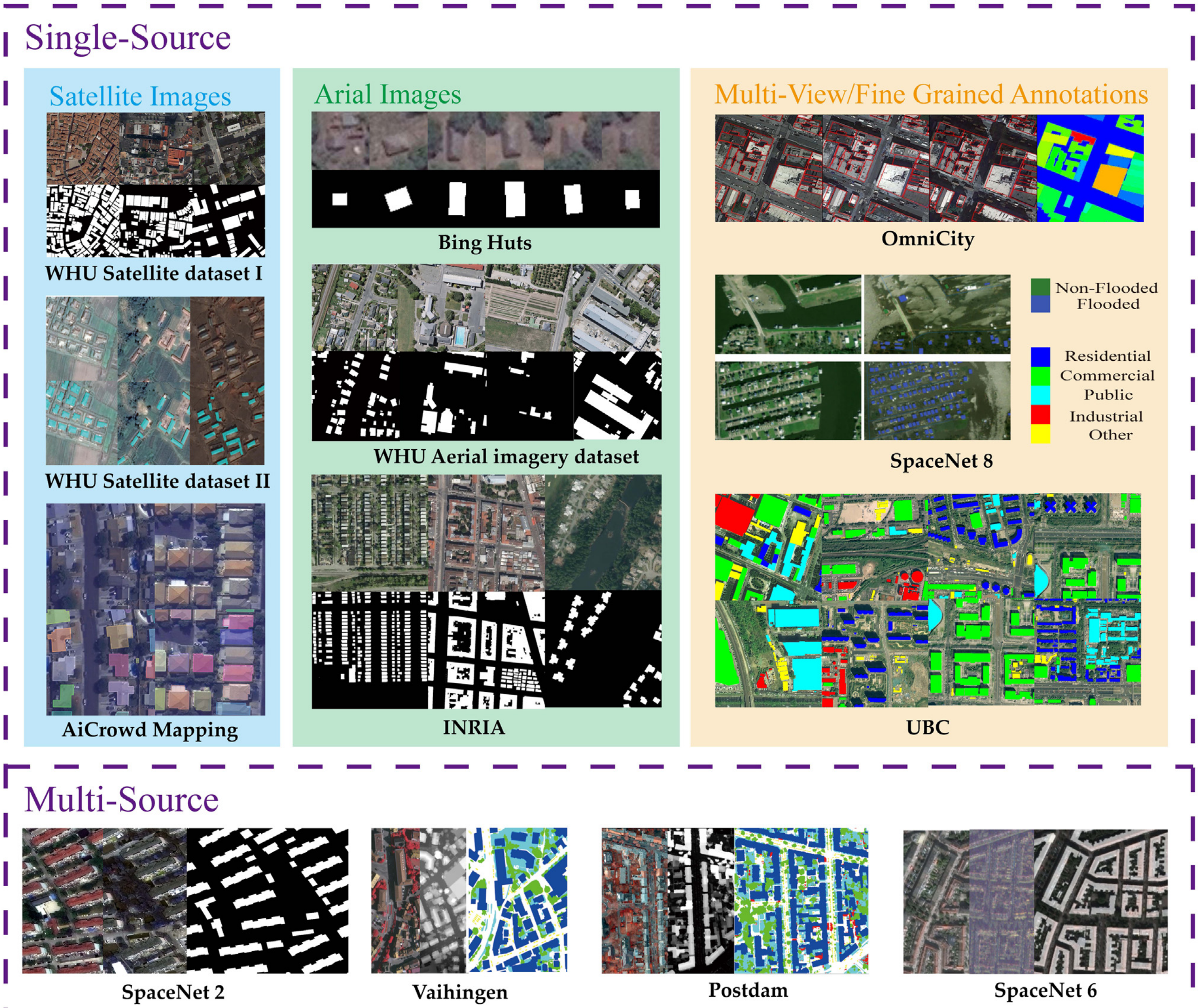

Fig. 12. Samples of public datasets for building extraction. *Source:* Some samples are adapted from the original papers Hänsch et al. (2022), Huang et al. (2022b).

6. Benchmark datasets, model performance and free productions

6.1. Publicly available datasets

Large-scale and high-quality labeled datasets are essential for model training and inference performance evaluation of DL-based building extraction methods. As the research on building extraction continues to evolve, many datasets have been constructed. These datasets show the trends of higher resolution, more diverse modalities, and richer annotations, bringing opportunities to develop more effective algorithms. In this section, a brief description of some representative publicly available datasets is given, including single-source datasets and multi-sources datasets (No non-publicly accessible datasets were involved). It should be noted that this survey began in 2023, and through the subsequent review and revision process, datasets published between 2013 and 2024 were collected. Given that the purpose of this section is to summarize relevant resources for reader convenience, only currently publicly accessible datasets are introduced and presented here. The key attributes of these datasets are listed in Table 7 chronologically. Some samples from the typical datasets are illustrated in Fig. 12. Due to space constraints, the access links to these datasets will be summarized at https://github.com/sxf118/Awesome-Building-Extraction (All links are accessible for dataset download, verified on 16, October, 2024.).

6.1.1. Single-source datasets

- **Massachusetts Building Dataset** (Mnih, 2013): This dataset utilizes the images provided by the state of Massachusetts as samples and the maps obtained from OpenStreetMap as labels.
- **INRIA Aerial Image Labeling** (Maggiori et al., 2017a): This dataset contains several high-resolution images covering areas with different population densities, *e.g.*, San Francisco and Lienz. It can provide rich patterns of building objects.
- **Bing Huts** (Marcos et al., 2018): Bing Huts dataset covers a rural area in Tanzania including 605 individual huts. The annotations are obtained from OpenStreetMap.
- **AiCrowd Mapping Challenge** (Web, 2018b): This dataset provides more than 34,000 satellite images tiles and MS-COCO format annotations of buildings.
- **OpenAI Dataset** (Web, 2018a): It is a part of 2018 OpenAI Tanzania Building Footprint Segmentation Challenge Competition, providing four types of building annotations: complete, incomplete, foundation, and none. These huge images are collected by Tanzanian drone pilots.

Table 7
The list of public datasets for building extraction literatures.

Datasets	Modality	#Class of Annot.	Image size	#Image	Resolution	Scene	Annot. type	Platform	MV/FGA	Year
Single-source										
Massachusetts Building Dataset (Mnih, 2013)	RGB	2	1500*1500	151	1 m	urban	raster	aerial	××	2013
INRIA Aerial Image Labeling (Maggiori et al., 2017a)	RGB	2	5000*5000	360	0.3 m	urban	raster	aerial	××	2017
Bing Huts (Marcos et al., 2018)	RGB	2	80*80	605	0.3 m	rural	raster	aerial	××	2018
AiCrowd Mapping Challenge (Web, 2018b)	RGB	2	300*300	401 755	/	urban	polygon	satellite	××	2018
OpenAI Dataset (Web, 2018a)	RGB	4	40000*40000	13	0.07 m	rural	polygon	aerial	×✓	2018
Vaihingen Buildings (Marcos et al., 2018)	NIRRG	2	512*512	168	0.09 m	urban	raster	aerial	××	2018
WHU Aerial Imagery Dataset (Ji et al., 2019)	RGB	2	512*512	8189	0.3 m	urban	raster	aerial	××	2019
WHU Satellite Dataset I (Ji et al., 2019)	RGB	2	512*512	204	0.3m-2.5 m	urban	raster	satellite	××	2019
WHU Satellite Dataset II (Ji et al., 2019)	RGB	2	512*512	17 388	2.7 m	various	polygon/raster	satellite	××	2019
SpaceNet7 (Van Etten et al., 2021)	RGB	2	1024*1024	2389	4 m	urban	polygon	satellite	××	2021
SDLED (Xia et al., 2021)	RGB	2	512*512	1908	0.522 m	urban	raster	/	××	2021
Sentinel-2 (Dixit et al., 2021)	NIR-RGB	2	128*128	2592	10 m	urban	raster	satellite	××	2021
SpaceNet8 (Hänsch et al., 2022)	RGB	5	1300*1300	881	0.3m-0.8 m	various	polygon	satellite	×✓	2022
UBCv1 (Huang et al., 2022b)	RGB	61	600*600	800	0.5m-0.8 m	urban	polygon	satellite	×✓	2022
BONAI (Wang et al., 2022b)	RGB	3	1024*1024	3300	0.3m/0.6 m	urban	polygon	aerial	×✓	2022
OmniCity (Li et al., 2022b)	RGB	2	512*512	108 600	0.3 m	urban	polygon/raster (height)	satellite	✓✓	2023
WHU-Mix (Luo et al., 2023; Wei et al., 2023)	RGB	2	various	51445(raster)/ 64387(polygon)	various	various	polygon/raster	aerial/satellite	××	2023
SMAB (Deng et al., 2023)	RGB	2	256*256	2125	2.2 m	mountainous area	polygon	satellite	××	2023
BFE-Set (Wang et al., 2023f)	NIR-RGB	3	512*512	10 485	0.8 m	counties	raster	satellite	×✓	2023
Multi-source										
ISPRS 2D Vaihingen (Rottensteiner et al., 2014)	NIRRG, LiDAR(DSM)	6	2500*2500	33	0.09 m	urban	raster	aerial	××	2014
ISPRS 2D Postdam (Rottensteiner et al., 2014)	NIRRG/RGB, LiDAR(DSM)	6	6000*6000	38	0.05 m	urban	raster	aerial	××	2014
USGS (Bradbury et al., 2016)	NIR-RGB,LiDAR	3	5000*5000	25	0.15m-0.3 m	urban	polygon	aerial	××	2016
SpaceNet1 (Etten et al., 2018)	8MSI, RGB	2	650*650RGB, 163*163MSI	/	1mMSI, 0.5mRGB	urban	polygon	satellite	××	2016
SpaceNet2 (Etten et al., 2018)	8MSI, PAN	2	650*650PAN, 163*163MSI	24 586	1.24mMSI, 0.3mPAN	urban	polygon	satellite	××	2018
SpaceNet4 (Weir et al., 2019)	8MSI,PAN, RGB, NIR	2	900*900	60 000	0.46m-1.67 m	various	polygon	satellite	✓×	2019
SpaceNet6 (Shermeyer et al., 2020)	PAN, 4MSI, SAR	2	900*900	3401(train set)	2mMSI,0.5mPAN, 0.5mSAR	port	polygon	aerial/satellite	××	2020
BISM (Yuan and Mohd Shafri, 2022)	NIRRG/RGB, LiDAR(DSM)	2	512*512	2496	0.3 m	urban	polygon/raster	aerial	××	2022
DFC23 (Claudio et al., 0000)	RGB, SAR, DSM	12 (Track 1), 2 (Track 2)	512*512	9767	0.5m/0.8mRGB, 1mSAR, 2mDSM	urban	polygon	satellite	×✓	2023
UBCv2 (Huang et al., 2023)	RGB, SAR	12	512*512	11 336	0.5m/0.8mRGB, 1mSAR	urban	polygon	satellite	×✓	2023

1 MV refers to multi-view, and FGA refers to fine grained annotations. 4MSI and 8MSI represent the multispectral image with four and eight channels respectively, and NIR refers to near-infrared.
2 Verified on 16, October, 2024, all datasets are available for download.

- **Vaihingen Buildings** (Marcos et al., 2018): This dataset provides 168 buildings obtained by processing the train set of the ISPRS 2D Vaihingen (Rottensteiner et al., 2014). It is widely used for evaluating the ACMs-based methods.
- **WHU Aerial Imagery Dataset** (Ji et al., 2019): The dataset collects aerial images from the New Zealand Land Information Services website. And Christchurch's building vector data are edited manually, containing more than 220,000 independent buildings.
- **WHU Satellite Dataset I** (Ji et al., 2019): WHU Satellite Dataset I consists of satellite images of several cities and manually delineated building labels. It is a suitable choice for evaluating the robustness of the model.
- **WHU Satellite Dataset II** (Ji et al., 2019): WHU Satellite Dataset II provides images of 6 areas in East Asia and contains more than 34,000 buildings that have similar architectural styles. The annotations are also manually delineated.
- **SpaceNet7** (Van Etten et al., 2021): SpaceNet 7 is also called Multi-Temporal Urban Development SpaceNet dataset. It contains images over dynamic areas for 24 months and annotates each building instance with a unique identifier.
- **SDLED** (Xia et al., 2021): The dataset provides images covering the Beijing area from Google Earth. Massive unlabeled samples are also available for designing the semi-supervised learning model.
- **Sentinel-2** (Dixit et al., 2021): This dataset collects Near-InRared (NIR), blue, green, and red bands images covering the areas of Delhi, Bengaluru, and Hyderabad. Different combinations of these four bands make up the three sub-datasets.
- **SpaceNet8** (Hänsch et al., 2022): The released part of this dataset consists of two areas: Germany after heavy rains in early July 2021 and Louisiana after Hurricane Ida end of August 2021. It can be exploited for building footprint extraction and flooded building detection.
- **UBCv1** (Huang et al., 2022b): The dataset release RGB data for urban areas of Beijing and Munich. Besides building footprint masks, its annotations consist of 5 coarse-grained classes and 25 fine-grained classes of building roofs, as well as 5 coarse-grained classes of building functionality/usage.
- **BONAI** (Wang et al., 2022b): This dataset is collected from Google Earth and Microsoft Virtual Earth. It provides building instances from off-nadir images with annotations of roofs, footprints, and corresponding offset vectors.

- **OmniCity** (Li et al., 2022b): OmniCity provides extensive satellite-level and street-level images. Specifically for building extraction, it contains multi-view satellite images with annotations of building footprint and height.
- **WHU-Mix** (Wei et al., 2023; Luo et al., 2023): WHU-Mix is obtained by selecting samples, correcting original labeling errors, and adding new samples on the basis of WHU datasets (Ji et al., 2019) and INRIA dataset (Maggiori et al., 2017a). Its larger scale and more variable samples are conducive to training more generalized models. There are two different versions of this dataset, raster version and vector version, with distinct differences in data composition and annotation format.
- **SMAB** (Deng et al., 2023): This dataset collects the coordinates of latitude and longitude for *Scattered Mountainous Area Buildings* (SMAB) with manual labels, which is valuable for researches such as mountain disaster emergency.
- **BFE-Set** (Wang et al., 2023f): BFE-Set is a large-scale, multi-spectral dataset using GaoFen-2 satellite imagery for fine-grained building extraction, categorizing buildings across 20 disaster-prone regions in China into three structure types: steel/reinforced concrete, masonry, and block stone.

6.1.2. Multi-source datasets

- **ISPRS 2D Vaihingen** (Rottensteiner et al., 2014): The dataset consists of DSM and orthorectified multispectral images, i.e., near-infrared, red, and green. Its original annotations include 6 classes, yet many researchers employed it by regarding non-building classes as one class for evaluating the building extraction models as well.
- **ISPRS 2D Postdam** (Rottensteiner et al., 2014): This dataset also contains DSM and orthorectified images, while the orthorectified images are organized in two combinations. Its ground truth categories are as the same ISPRS 2D Vaihingen.
- **USGS** (Bradbury et al., 2016): The dataset released by *United States Geological Survey* (USGS) includes aerial images and LiDAR data, providing annotations of building footprints and road polylines across 25 locations in 9 U.S. cities.
- **SpaceNet1** (Etten et al., 2018): The images in this dataset are 8-band multispectral and pan-sharpened RGB images, covering more than 2500 kilometers in areas in Rio de Janeiro.
- **SpaceNet2** (Etten et al., 2018): SpaceNet 2 includes satellite images of four areas: Las Vegas, Paris, Shanghai, and Khartoum. It also provides pan-sharpened RGB and 8-band MSI data.
- **SpaceNet4** (Weir et al., 2019): This dataset contains multi-view overhead images of Atlanta. For each view combination, an RGB image, a PAN image, a NIR image, and an 8-band MSI are included.
- **SpaceNet6** (Shermeyer et al., 2020): SpaceNet 6 is also known as *Multi-Sensor All Weather Mapping* (MSAW) dataset, which covers the port of Rotterdam. It provides both quad-polarized X-band SAR images and optical images for training and only SAR image for testing.
- **BISM** (Yuan and Mohd Shafri, 2022): It is created for building instance segmentation task, consisting of MSI and LiDAR data that cover an area of 60 km2 in Boston, USA.
- **DFC23** (Claudio et al., 0000): This dataset contains optical images and SAR images collected from the SuperView-1, Gaofen-2 and Gaofen-3 satellites. Nearly 300k building instances distributed across seventeen cities are labeled with fine grained roof types or building height.
- **UBCv2** (Huang et al., 2023): This dataset is an extension of the Tarck 1 of DFC23 (Claudio et al., 0000), providing optical images and roof type annotations for three additional cities.

6.1.3. Remark

The annotations in most of the above datasets are distinguished as buildings and non-buildings. To be noticed, many land cover mapping datasets that contain labels of building instances are also valuable for building extraction, such as Web (2017), Xu et al. (2019), Roscher et al. (2020), Wang et al. (2021a).

Based on Table 7, the following discussion covers two aspects: key findings and insights from the table, and the differences among these datasets along with their impact on building extraction results.

①Key Findings. Firstly, recent advances in imaging technology have enabled datasets from either aerial or satellite platforms to achieve sub-meter resolution. These high-resolution images enhance levels of detail, facilitating the extraction of finer building masks and supporting applications such as urban planning and 3D urban reconstruction. However, it is important to note that increased resolution also requires greater computing and hardware storage capacity, and the use of image cropping or downsampling methods may result in the loss of potential features. Therefore, more efficient algorithms are necessary to manage these demands.

Secondly, the latest datasets are characterized by multi-source images or more detailed annotations. On one hand, these datasets effectively promote the development of building extraction methods based on multi-source data and enable more comprehensive evaluation of models. On the other hand, they also present new tasks and challenges such as how to learn perspective deviation to accurately extract building footprints (Wang et al., 2022b), how to combine satellite and street view images for multi-level city understanding (Li et al., 2022b), and how to conduct fine-grained visual feature learning and multi-task joint training based on multi-modal data (Claudio et al., 0000).

Thirdly, most datasets focus on urban scenes or mixed scenes dominated by urban areas. This is primarily due to the urgent application needs in urban regions, such as urban planning and disaster response. And urban buildings are more complex and diverse in form, posing great research and algorithmic challenges.

Lastly, it should be noted that the scale of multi-source datasets are generally smaller than single-source datasets, resulting in a greater richness of building extraction methods based on optical images compared to those based on multi-source data. However, the complementary advantages of multi-source data are irreplaceable. Comprehensive utilization of multi-source data should be the mainstream demand in the future. Therefore, researchers should pay more attention to the imbalance of available data quantity and aim to address this in future studies.

② Differences and Impact. First, the spatial resolution of images in these datasets varies significantly, ranging from 0.05 m in the ISPRS 2D Potsdam dataset to 10 m in the Sentinel-2 dataset. The ultra-high-resolution images mainly come from aerial platforms, while high-resolution images are primarily from advanced satellite platforms. Low-resolution images can lead to blurred building boundaries and loss of detail, particularly for small buildings, while extremely high-resolution images increase computational costs and make it difficult for models to capture global dependencies. To address these challenges, techniques such as multi-scale learning, super-resolution, and ultra-high-resolution image processing should be flexibly employed based on the resolution characteristics of the data to improve building extraction performance.

Second, the types of sensors used for imaging in these datasets differ. The differences in the imaging mechanisms of sensor result in varying characteristics. PAN and RGB images provide texture information but are highly dependent on lighting conditions. MSI capture spectral reflectance characteristics of building surfaces but have lower spatial resolution and require significant data processing. LiDAR provides elevation information but lacks color and texture features. SAR is suitable for imaging in harsh environments but is prone to speckle noise and geometric distortions. To leverage the advantages of multiple sensors, multi-source fusion is typically needed to improve the accuracy and robustness of building extraction.

Third, the spectral bands involved in these datasets also vary. Most datasets provide RGB bands, some include near-infrared bands, and certain SpaceNet datasets offer four-channel or eight-channel MSI. The RGB bands provide intuitive color and texture information, suitable for delineating building boundaries. NIR bands enhance the model's ability to distinguish buildings from vegetation. MSI data, by combining information from multiple bands, improves the differentiation of building materials and spectral details. In the process of constructing building extraction datasets, preprocessing techniques like pan-sharpening are often employed to obtain high-quality inputs for improving extraction results.

Finally, scene differences in these datasets also affect building extraction accuracy and model optimization difficulty. In urban scenes, buildings are densely packed and highly varied in form, requiring careful attention to confusion between neighboring buildings. In mixed scenes dominated by urban areas, the imbalance in building density and types can lead to model biases favoring dominant patterns, making model optimization more challenging. Additionally, when applying techniques like semi-supervised learning or domain adaptation, attention must be paid to the scene differences between unlabeled data and labeled data to ensure efficient and effective data utilization.

6.2. Evaluation metrics

① Pixel-Based. In the relevant literature, the following five pixel-based metrics are widely used to evaluate the performance of building extraction methods, i.e., *Overall Accuracy* (OA), Recall, Precision, F1 score, and *Intersection over Union* (IoU). The specific definition is as follows,

$$OA = \frac{TP + TN}{TP + FP + TN + FN} \tag{3}$$

$$Precision = \frac{TP}{TP + FP} \tag{4}$$

$$Recall = \frac{TP}{TP + FN} \tag{5}$$

$$F_1 = 2 \times \frac{precison \times recall}{precison + recall} \tag{6}$$

$$IoU = \frac{target \cap detected}{target \cup detected} = \frac{TP}{TP + FP + FN} \tag{7}$$

where TP is the number of positive samples that are correctly classified, FP is the number of positive samples that are misclassified, TN is the number of negative samples that are correctly identified, and FN is the number of negative samples that are wrongly identified. For the building extraction task, the positive sample is the building pixel, and the negative sample is the non-building pixel.

OA gives the proportion of all correctly classified samples in the data set. Recall evaluates the completeness of the model's prediction. Precision measures the accuracy of the model. F1 is the harmonic average of Recall and Precision. And IoU measures the localization accuracy of the model.

② Region-Based. There are some other region-based metrics, such as *Boundary F-score* (BoundF), *Weighted Coverage* (WCov), and *Rotation Error* (RE) (Hatamizadeh et al., 2020; Huang et al., 2022c):

$$Bound\,F = \frac{1}{T} \sum_{i=1}^{T} \frac{2 P_i R_i}{P_i + R_i} \tag{8}$$

where P_i and R_i are precision and recall respectively, which only take boundary pixels into account. T is a threshold value to control how many boundary pixels are used.

$$WCov(S_P, S_G) = \frac{1}{N} \sum_{j=1}^{|S_G|} |r_j^{S_G}| \max_{k=1 \ldots |S_X|} IoU(r_k^{S_P}, r_j^{S_G}) \tag{9}$$

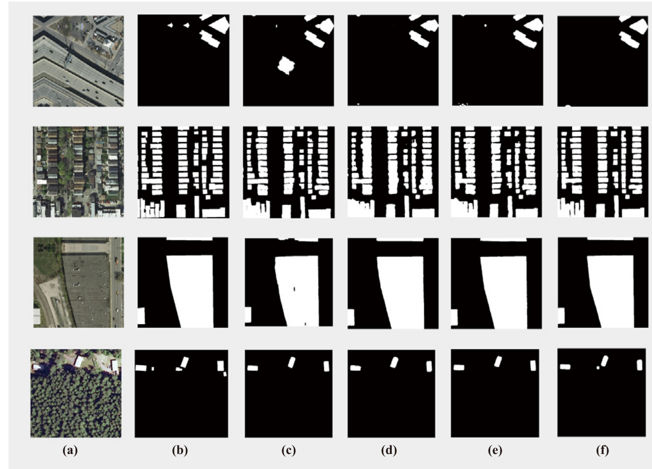

Fig. 13. Visual comparison of some raster output methods for building extraction on INRIA dataset: (a) image, (b) ground truth, (c) U-Net, (d) DeepLab, (e) BuildFormer, (f) UANet.

where $S_X = \{r_1^{S_P} \ldots r_{|S_P|}^{S_P}\}$, $r_1^{S_P}$ is a predicted polygon region, $|r_j^{S_P}|$ is the number of pixels in $r_1^{S_P}$, and similar to ground truth region set S_G. This metric reflects the ratio between the predicted and the ground truth polygon region.

$$RE = |O_p - O_g| \tag{10}$$

where O_p and O_g are orientations of the predicted polygon and that of the ground truth polygon, respectively. The orientation is measured by the angle formed between the x-axis and the major axis of an ellipse that has the same second moments as the given polygon.

③ Others. Besides the metrics above, *Average Precision* (AP) (Xu et al., 2022a) (commonly used for evaluating methods based on the instance segmentation paradigm.), *Polygon Similarity* (PolySim) (Zhu et al., 2022), *Hausdorff Distance* (HD) and *Structural SIMilarity* (SSIM) (Zhou et al., 2022) are also can be reasonable choices to evaluate the performance of building extraction methods.

6.3. Performance comparison and discussion

By investigating the recent relevant literatures, few datasets are found to be widely accepted by the community to evaluate models. Many researchers test their methods on both the public and the private datasets. Here, three frequently used datasets are selected: INRIA dataset, Vaihingen Building dataset and ISPRS 2D Vaihingen dataset. The performance achieved by recent advanced methods is reported in Tables 8, 9 and 10. To comprehensively showcase the performance of various SOTA building extraction methods, the studies summarized here are from 2018 to 2024, with the majority focusing on work from 2022 and later. All the results are consistent with the original papers, where the experiments were conducted in different settings. The key experimental settings are reported in these tables as well. Moreover, to gain a more intuitive understanding of the performance of these methods, we trained several open-source general image segmentation models (U-Net (Ronneberger et al., 2015) and DeepLab Chen et al., 2018b) and building extraction models (BuildFormer (Wang et al., 2022a), UANet (Li et al., 2024a), DARNet (Cheng et al., 2019), and CVNet (Xu et al., 2022b)) from scratch based on the settings provided in the original literature. Depending on the different output formats of these models, their visualization results on INRIA dataset and Vaihingen Building dataset are displayed in Fig. 13 and Fig. 14, respectively. In addition, python library *ptflops* (Web, 2022a) is used to estimate the computations and the number of parameters of recent

Table 8
Performance comparison of some building extraction methods on INRIA dataset.

Methods	OA (%)	Recall (%)	Precision (%)	F1 (%)	IoU (%)	Training epochs	Initial learning rate	Training-test ratio	Output type	Publication	Year
UANet (Li et al., 2024a)	/	89.52	92.04	90.76	83.08	/	5e−4	67:5	raster	IEEE TGRS	2024
HD-Net (Li et al., 2024b)	97.32	89.26	91.10	90.17	82.10	approximately 100 epochs	1e−3	2463:500	raster	ISPRS JPRS	2024
SDSC-UNet (Zhang et al., 2023f)	/	89.92	91.52	90.71	83.01	105	/	67:5	raster	IEEE GRSL	2023
TopDiG (Yang et al., 2023b)	94.70	/	/	91.32	84.56	/	/	17:1	polygon	CVPR	2023
PolyBuilding (Hu et al., 2023a)	/	/	/	/	81.20	200	2e−4	3:1	polygon	ISPRS JPRS	2023
C3Net (Gong et al., 2023)	/	85.62	87.00	86.30	76.04	/	1e−4	2025:891	raster	KBS	2023
BOMSC-Net (Zhou et al., 2022)	95.31	87.93	87.58	87.75	78.18	200	1e−4	67:5	raster	IEEE TGRS	2022
B3SM (Lee et al., 2022)	98.74	86.76	86.08	86.42	76.08	200	1e−3	4:1	raster	IEEE TGRS	2022
U-Net-AFM (Li et al., 2022c)	/	/	/	86.68	76.49	100	1e−5	67:5	raster	IEEE TGRS	2022
RAPNet (Tian et al., 2022)	96.74	/	/	/	76.78	/	1e−4	67:5	raster	IEEE GRSL	2022
ASF-Net (Chen et al., 2022b)	94.7	/	/	96.7	80.2	10K iterations	1e−3	1:1	raster	IEEE TGRS	2022
CGSANet (Chen et al., 2022e)	96.91	88.07	89.32	88.69	79.68	100	/	67:5	raster	IEEE JSTARS	2022
CVNet (Xu et al., 2022b)	/	/	/	/	77.61	250	8e−3	4:1	polygon	CVPR	2022
ISDNet (Guo et al., 2022b)	/	/	/	/	74.23	10K iterations	1e−3	17:3	raster	CVPR	2022
BuildFormer (Wang et al., 2022a)	/	88.81	90.75	89.77	81.44	105	5e−4	67:5	raster	IEEE TGRS	2022
ASLNet (Ding et al., 2022)	97.15	86.85	90.00	88.27	79.30	50	/	67:5	raster	IEEE TIP	2022
DS-Net (Zhang et al., 2022a)	96.46	/	/	/	79.54	60K iterations	1e−4	67:5	raster	IEEE TNNLS	2022
CBR-Net (Guo et al., 2022a)	/	89.2	89.93	89.56	81.1	dynamic	1e−3	2441:625	raster	ISPRS JPRS	2022
FFL (Girard et al., 2021)	/	/	/	/	74.8	300 iterations	1e−1	3:1	polygon	CVPR	2021
STT (Chen et al., 2021)	96.59	/	/	87.99	79.42	300	1e−2	4:1	raster	Remote Sensing	2021
MA-FCN (Wei et al., 2020)	/	85.57	87.57	/	76.33	/	1e−4	3:1	raster	IEEE TGRS	2020

Table 9
Performance comparison of some building extraction methods on ISPRS 2D Vaihingen dataset.

Methods	OA (%)	Recall (%)	Precision (%)	F1 (%)	IoU (%)	Training epochs	Initial learning rate	Training-test ratio	Output type	Publication	Year
optical											
OEC-RNN (Huang et al., 2022c)	/	/	/	/	90.02	/	1e−4	4:1	polygon	IEEE TGRS	2022
optical and DSM											
ACMFNet (Chen et al., 2024a)	/	/	/	89.76	94.60	105	1e−3	16:17 (patches)	raster	IEEE TGRS	2024
Luo et al. (Luo et al., 2024)	97.64	/	/	95.72	91.80	/	1.5e−4	7:3	raster	IEEE TGRS	2024
CMGFNet (Hosseinpour et al., 2022)	96.91	/	/	95.96	88.84	dynamic	1e−3	4:1	raster	ISPRS JPRS	2022
HAFNet (Zhang et al., 2020)	97.04	/	/	98.17	87.32	50	multiple	7:3	raster	Remote Sensing	2020
Res-U-Net (Xu et al., 2018)	97.71	94.12	96.21	95.15	/	100	1e−3	4:1	raster	Remote Sensing	2018
V-FuseNet (Audebert et al., 2018)	/	/	/	94.4	/	/	1e−2	/	raster	ISPRS JPRS	2018

Table 10
Performance comparison of some building extraction methods on Vaihingen Building dataset.

Methods	BoundF (%)	WCov (%)	F1 (%)	mean IoU (%)	Training epochs	Initial learning rate	Training-test ratio	Output type	Publication	Year
CVNet (Xu et al., 2022b)	81.76	90.46	/	90.43	250	8e−3	25:17	polygon	CVPR	2022
APGA (Zhu et al., 2022)	/	90.40	/	90.40	/	1e−2	25:17	polygon	IEEE TGRS	2022
ACDRNet (Gur et al., 2020)	79.19	89.03	95.62	91.74	/	1e−3	25:17	polygon	ICLR	2020
TDAC (Hatamizadeh et al., 2020)	78.12	90.54	/	89.16	/	1e−3	25:17	polygon	ECCV	2020
DARNet (Cheng et al., 2019)	75.91	88.16	/	88.24	/	4e−5	25:17	polygon	CVPR	2019

open-source models with input size of 224 × 224 pixels, which are presented together with IoU on INRIA dataset in Fig. 15.

As shown in Tables 8 and 9, almost all methods achieve an OA of more than 95%, while other metrics are not as high. This indicates that OA is prone to an over-optimistic assessment of the model due to the sample imbalance between building and non-building. For example, assuming 90% of the samples are non-building and 10% are building in a dataset, the model only needs to predict all samples as non-building to reach 90% high accuracy, although none of the positive samples is correctly predicted. Therefore, F1 and IoU are more reasonable metrics.

For the building extraction task, besides high-level semantic information, low-level detailed features are also vital. Some SOTA methods in Table 8 can be attributed to this. For instance, DS-Net (Zhang et al., 2022a) consists of two complementary branches, the local branch and the global branch, that work in a complementary manner with different fields of view on the input image. BuildFormer (Wang et al., 2022a) utilizes the transformer-based dual-branch architecture to learn more discriminative features and merges low-level and high-level features with a contextual aggregation module. Furthermore, the combination of low-level and high-level features is significant for damage evaluation under actual disaster scenarios (Xie et al., 2022). To identify the damaged buildings, it is necessary to properly abstract their semantic features, and to accurately assess the damage degree, it is necessary to grasp the texture changes of the damaged buildings.

As evidenced by the varying proportions listed in the training-test ratio column of Table 8, there is no universally accepted standard for the division of training and test sets for the same dataset across different work. Even when the division ratio is consistent as Table 10 indicates, the selection of samples is often random. Such inconsistencies in dataset partitioning make it impossible to directly and impartially compare different methods. Researchers are often required to expend repetitive works to evaluate model performance under their own dataset partitioning. To overcome this challenge, it is recommended that researchers make their source code and detailed dataset partitioning publicly available to enable consistent and fair comparisons and promote technical advances in building extraction.

Furthermore, considering the output types of methods in recent three years on the three datasets, the majority of them output in raster format. This output format aligns with the general vision paradigm and facilitates rapid technological progress. In addition, prediction results in polygon output have clear boundaries and are readily applicable to downstream GIS applications. As such, relevant technical research of polygon output methods is also continuously developing.

For the visualization of results, as shown in Fig. 13, mainstream DL-based segmentation methods generally achieve satisfactory extraction performance. However, the models still tend to misclassify or confuse densely distributed building boundary areas and very small buildings. In Fig. 14, it can be observed that current mainstream methods for direct polygon prediction can generally fit building contours, but some deviations still may occur at certain corner points.

In addition, by combining Table 8, Tables 9 and 10, it can be concluded that the recent research applying multi-source data is not abundant enough. On the one hand, this is consistent with the above analysis of the datasets. On the other hand, there are many combinations of multi-source data and strong specificity of methods for different combinations, resulting in relatively scattered studies.

Lastly, what is supposed to be noted in Fig. 15 is that STT (Chen et al., 2021) performs satisfactorily with the least amount of calculations and parameters, achieving a great trade-off between model accuracy and complexity. From the perspective of practical application requirements, this trade-off is necessary. For example, in disaster emergency scenarios, only rapid and accurate extraction of building footprint information can provide support for subsequent decision-making. Looking back at STT, one of the key ideas of it is adopting a sparse sampling strategy to optimize features. This inspires researchers

Fig. 14. Visual comparison of some polygonal output methods for building extraction on Vaihingen Building dataset: (a) image, (b) ground truth, (c) DARNet, (d) CVNet.

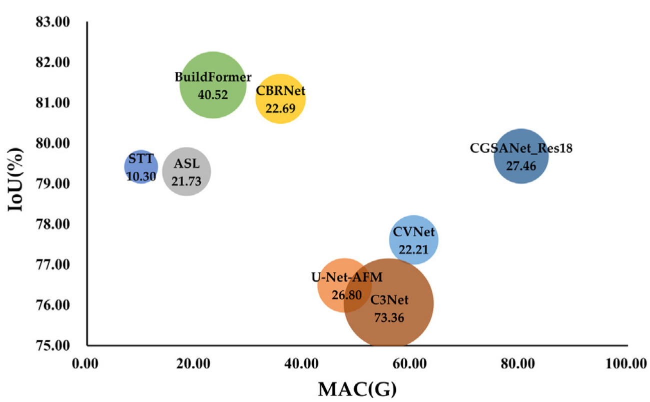

Fig. 15. Comparison of some open-source methods in terms of computation, number of parameters, and IoU on INRIA dataset. MAC refers to Multiply–Accumulate Operations, and the value on the bubble is the number of parameters (M), which also represents the bubble size.

that it is vital to pay more attention to the strong semantic sparsity of remote sensing images for improving accuracy and reducing hardware overhead simultaneously.

6.4. Free-available productions

With the advancement of DL, researchers and institutions have recently begun developing large-scale, high-precision, free-available building data products. These building footprints products are largely developed using advanced DL models combined with carefully designed preprocessing and post-processing techniques, providing valuable resources and references for both academic research and practical applications. To better assist readers in understanding these products, a brief summary is provided as following, and a comparison of their key attributes is shown in Table 11. The time range considered here is focused on 2021 and beyond, as earlier products primarily emphasized impervious surfaces (Huang et al., 2021) or human settlement footprints (Marconcini et al., 2020) rather than building footprints.

- **Google Open Buildings** (Web, 2021): The Google Open Buildings contains 1.8B building detections across 58M square kilometers, providing building footprints and confidence scores for each building.
- **Microsoft** (Web, 2022b): This product, based on Microsoft's Bing Maps imagery, provides 1.4B building footprints worldwide (excluding China) from 2014 to 2024 and is continually updated on an irregular basis.
- **90-cities** (Zhang et al., 2022c): This production provides vectorized rooftop data for 90 cities in China, generated using high-resolution open-access remote sensing imagery.
- **Google Open Buildings Temporal** (Web, 2023a): The Google Open Buildings 2.5D Temporal Database provides annual data from 2016 to 2023, including building presence, counts, and heights, using Sentinel-2 satellite imagery.
- **EUBUCCO** (Milojevic-Dupont et al., 2023): This database contains building footprints for over 200M buildings with attributes like building type, height, and construction year, compiled from 50 open government datasets and OpenStreetMap.
- **CBRA** (Liu et al., 2023b): *Building Rooftop Area* (BRA) production is multi-annual and full-coverage in China, produced by deep segmentation model with super-resolution method using Sentinel-2 imagery.
- **GABLE** (Sun et al., 2024): This production is a nation-scale fine-grained building database derived from Beijing-3 satellite imagery, providing compact rooftop polygons, geometric categories for 12 rooftop types, and height values for individual buildings.
- **East Asian Buildings** (Shi et al., 2024b): It provides comprehensive building footprints for 281,093,433 buildings across 2897 cities in 5 East Asian countries, with satisfactory evaluation performance.
- **CBF** (Huang et al., 2024): This is China's first sub-meter building footprint database, generated using a semi-supervised training approach to utilize the incomplete information from OSM.

7. Prospects

Although recent works have achieved gratifying results, there are still some issues that need to be further explored. In this section, based on the review of previous work and recent advances in related fields, the future directions are prospected from the following three aspects.

7.1. Practical application oriented algorithm design

As discussed above, by utilizing multi-scale, multi-task, and geometry prior strategies to extract more discriminative features, the accuracy has reached a satisfactory level, leading to the emergence of many free building footprint products (Web, 2021; Huang et al., 2024). While higher accuracy metric values on valuation dataset are desirable, they do not sufficiently meet the diverse needs of practical applications. Remote sensing data in real-world applications presents additional challenges such as larger image sizes, and variations in imaging angles. Furthermore, there are limitations imposed by algorithm running environments and the need for fine-grained analysis, which cannot be fully addressed by improvements in model accuracy alone. These challenges deserve further exploration in the future, with potential novel technologies applied to open products.

7.1.1. Mitigation of the imbalance between foreground and background

The quantity imbalance between buildings and non-buildings harms the optimization process. To alleviate this problem, great efforts have been made for raster output methods, such as co-segmentation approach (Li et al., 2022d). However, adaptive responding to the samples distribution pattern remains a challenging problem. In addition, for polygon output methods that are more convenient for practical applications, the foreground elements (such as the corners, the edges

Table 11
The list of free-available productions of building footprints.

Product	Resolution	Areas	Time span	Release year
Google Open Buildings (Web, 2021)	0.5 m	Africa, South Asia, South-East Asia, Latin America and the Caribbean	/	2021
Microsoft (Web, 2022b)	<1 m	Global (Without China)	2014–2024 (Updating)	2022
90-cities (Zhang et al., 2022c)	1 m	China	2020	2022
Google Open Buildings Temporal (Web, 2023a)	4 m	Africa, South Asia, South-East Asia, Latin America and the Caribbean	2016–2023	2023
EUBUCCO (Milojevic-Dupont et al., 2023)	/	7 European Union countries and Switzerland	/	2023
CBRA (Liu et al., 2023b)	2.5 m	China	2016–2021	2023
GABLE (Sun et al., 2024)	0.5m-0.8 m	China	2023	2024
East Asian Buildings (Shi et al., 2024b)	0.5 m	China, Japan, South Korea, North Korea, and Mongolia.	2020–2023	2024
CBF (Huang et al., 2024)	0.5 m	China	2019	2024

of buildings) are further drastically reduced, resulting a more urgent demand for the algorithm of foreground and background balance learning.

7.1.2. Deal with the view bias

The vast majority of existing building extraction models and datasets are based on the assumption that the footprint and roof of a building are highly overlapped. But due to the change of imaging view, this assumption is not always true. This deviation limits the accuracy of footprint extraction and impairs confidence in the 3D reconstruction of buildings (Wang et al., 2022b). How to correlate the building's feature patterns under different image views? How to effectively learn the offset between roof and footprint? These questions are worthy of further exploration.

7.1.3. Lightweight and hardware-friendly algorithm

Traditional remote sensing data interpretation adopts the process of downloading data from the aero or satellite remote sensing platforms and analyzing them on the ground center. This process often takes several hours to complete and is far from reaching the requirements of highly responsive tasks. To this end, researchers are increasingly focused on intelligent in-orbit data processing, that is, the space-borne platform transmits critical information after processing (Ziaja et al., 2021; Zhang et al., 2023e). Considering the limited resources on the space-borne platform, the building extraction algorithm should be lightweight and hardware-friendly. Moreover, to full use of the information of each terminal, the algorithm should be distributed joint trainable (Wang et al., 2023a). However, most of the existing algorithms cannot meet these requirements. Thus related researches worth more attention.

7.1.4. Combination of footprints extraction and attribute analysis

The current state-of-the-art building extraction methods are limited to predicting only the footprint of the buildings. But for specific applications such as urban planning and GIS maintenance, the footprint information alone is far from enough. If the model can extract buildings and give building attributes at the same time, it will greatly enrich its application scenarios. For example, Mao et al. (2023a) perform building extraction and elevation estimation separately from the single view image using different non-shared network to support the 3D reconstruction. However, due to the information source from a single perspective, the analysis of building attributes is limited to elevation. With the assistance of street-view images, Workman et al. (2022) predict the height and age of the building based on the street panorama, while Kang et al. predict the age and function of the building (Kang et al., 2018). Moreover, He et al. (2023) design a generative network to extract building footprints and multiple geometric attributes with the supervision of globally sociogeometric features. Nevertheless, street-view images are not always available, and social features generally lag behind the image data in time phase. Thus, how to use only single-view images to accurately extract and analyze buildings in multiple aspects simultaneously is a very challenging but valuable task.

7.2. Joint analysis of multi-source data

Multi-source data can provide complementary features, helping to understand the building objects at a comprehensive level. However, real-world applications scenarios present numerous challenges, including but not limited to the constant enrichment of modality, unbalanced modal advantages, and the need for registration prior to use. So there are still a lot of work needed to do.

7.2.1. Flexible expansion of input

Most of the existing fusion frameworks can handle only two fixed types of data. However, the types and the combinations of multi-source data available often vary for different times and regions, which poses a challenge for existing fusion methods. And with the development of sensor technology, the data such as video and night light data are emerging, which requires the development of algorithms that can process multiple data at the same time. Although synthetic multi-modal data can be used to construct various data combinations to aid model training (Fuentes Reyes et al., 2023), maintaining models trained on all possible combinations to handle potential modality missing is too costly and impractical. Thus a framework that can flexibly process multiple modals is desirable. Recently, Zhang et al. (2023c) propose arbitrary-modal semantic segmentation paradigm, providing a reasonable solution to this problem in driving scenario. Further investigation is required on how to apply this paradigm in remote sensing scenarios, given the unique characteristics of remote sensing images and buildings.

7.2.2. Imbalance in multi-source data

The imbalance in multi-source data is manifested in task advantage and is dependent upon scenarios. For example, optical images offer greater advantages in well-lit environments, while SAR images are more effective in poor weather conditions. Given the inherent complexity and variability of real-world scenarios, the presence of such imbalance is likely to result in the sub-optimal performance of the model. This issue often requires explicit intervention in the training process to alleviate (Peng et al., 2022). Recently, Shi et al. (2024a) introduce a uni-modal feature enhancement framework for modal-imbalance in building extraction task. Although extraction performance has improved significantly, there is still room for enhancement in handling representation degradation caused by imbalances in highly noisy conditions, warranting further exploration of related methods in the future.

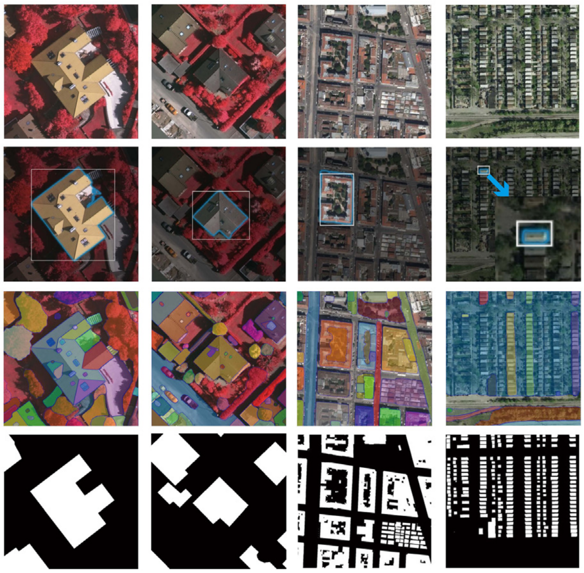

Fig. 16. The visualization results of SAM on some samples of Vaihingen Buildings datasets and INRIA datasets: (a) input images, (b) output of SAM with the bounding box prompt, (c) output of SAM with the "everything" prompt, (d) ground truth.

7.2.3. Unpaired multi-source image learning

The current multi-source data-based building extraction methods assume that there is a pixel-to-pixel correspondence between image pairs, which necessitates time-consuming and labor-intensive cross-modal image registration. Despite registration efforts, deviations may still occur, making it difficult to ensure pixel-level correspondence. In real-world applications, it is not guaranteed that the available modalities are always consistent with those during the training phase. Thus the exploration of methods for building extraction from unpaired multi-source data is of great practical significance. Several studies confirm the feasibility of implicit learning from unpaired data by generative adversarial (Xia et al., 2022), knowledge distillation (Kang et al., 2022b), or knowledge retention (Zheng et al., 2021b). However, there is still a gap between the performance of models obtained by these methods and traditional supervised learning. Related researches remain to be carried out.

7.3. Foundation models and large-scale training

With some excellent work from the computer vision community applied, the performance of building extraction models has improved significantly. Thus it is vital to keep abreast of the latest developments in computer vision research to further advance building extraction techniques. Noting that there has been considerable attention on large models and large-scale training in the computer vision community in recent time, some related studies are discussed here, to provide insights into the future of building extraction techniques at a more general level.

7.3.1. Improvement with foundation vision models

The *Segment Anything Model* (SAM) (Kirillov et al., 2023) is a representative foundation vision model that can perform promptable segmentation. It can receive prompts like points, bounding boxes, masks or text as inputs, and then outputs object masks. Its amazing zero-shot segmentation ability has recently garnered significant attention from researchers and practitioners alike, and it also brings important implications for various other vision tasks. However, its effectiveness in the specific task of building extraction has not been thoroughly reported. To evaluate SAM's performance on building extraction, we conduct some simple experiments using the official online demo, as shown in Fig. 16. Upon examining the first two columns, it is evident that for images with prominent buildings, SAM produces accurate boundaries with the bounding box prompt, and it exhibits a propensity for over-segmentation with the "everything" prompt (find all the objects automatically), such as segmenting building roofs into multiple blocks or dividing roof windows. From the analysis of the results shown in the last two columns, it is apparent that SAM's performance on building extraction is not always satisfactory, especially when dealing with buildings of complex shapes or densely distributed building groups. Specifically, when with the bounding box prompt, SAM misclassifies the trees inside a building. With the "everything" prompt, SAM segments a set of individual buildings as a whole. For quantitative results on building extraction task, Chen et al. (2023b) provides the results of SAM combined with different segmentation heads, classification head and detector on WHU Aerial dataset (Ji et al., 2019), which shows consistency with our visual analysis. To further validate the potential of the foundation model for building extraction tasks, two popular parameter-efficient fine-tuning methods are selected to test the fine-tuning performance of SAM (ViT_b) on the INRIA dataset. The quantitative results are shown in Table 12, and the visual results are presented in Fig. 17. It is evident that even only with basic parameter-efficient fine-tuning, SAM demonstrates highly competitive performance compared to several specialized methods developed in recent years.

Table 12
Performance comparison of SAM with some parameter-efficient fine-tuning methods for building extraction on INRIA dataset.

Methods	Recall	Precision	F1	IoU
Adapter (Houlsby et al., 2019)	82.42	91.38	86.67	76.47
LoRA (Hu et al., 2022)	83.78	92.24	87.81	78.26

Based on the above discussion and some recent researches, to leverage the power of large visual models to facilitate the development of building extraction methods, the following perspectives deserve attention. The first is the improvement of the data annotation process. Some studies point out that SAM has impressive automatic/semi-automatic annotation capabilities (Sun et al., 2023; Wang et al., 2023d). Therefore, it is of great significance to develop more efficient data annotation pipeline using large visual models to promote the development of large-scale and fine-grained building-related datasets. The second is to design efficient algorithms to mobilize the potential of large models on the building extraction task. Adapter (Chen et al., 2023d) is demonstrated to improve SAM's performance on specific segmentation tasks, such as camouflage segmentation and medical image segmentation. How to design a adapter according to the characteristics of the building extraction task to adjust SAM's segmentation granularity while improving its performance in dense and complex scenes? This needs to be further explored. In more recent research, prompt learning (Chen et al., 2023b) has been demonstrated to enhance the semantic instance segmentation capability of the SAM when applied to remote sensing images, providing another perspective to adapt SAM for building extraction. The third is to combine large visual models with multi-source data. Although the prevailing view is to use SAM to process RGB images, SAM has also been confirmed to be available for thermal infrared images (Chen and Bai, 2023). Does this generalization still hold for multi-source remote sensing data, and how can large models be better combined with multi-source data? These issues remain to be investigated.

7.3.2. Remote sensing pre-training

To obtain a large vision model that takes advantage of prior knowledge and performs well on specific tasks, the popular training paradigm is to initialize with model weights pre-trained on large-scale datasets, and then fine-tune on specific small-scale datasets. Intuitively, compared with the model pre-trained on ImageNet (Deng et al., 2009),

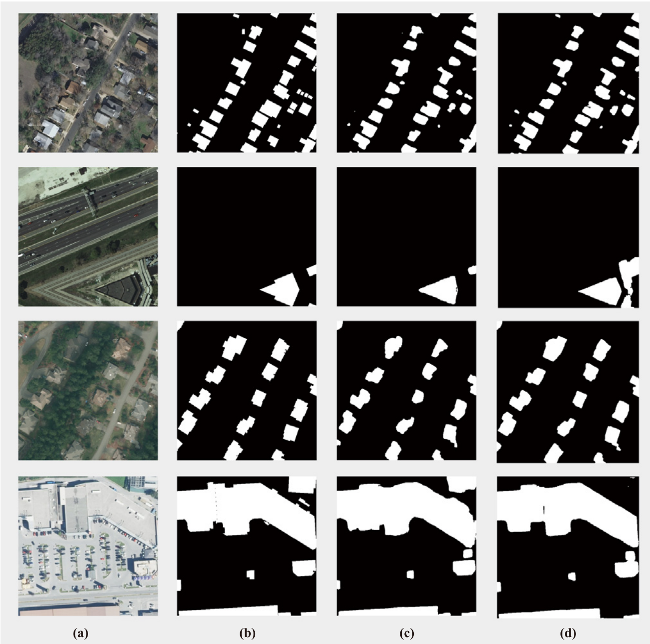

Fig. 17. Visual comparison of SAM with two parameter-efficient fine-tuning methods for building extraction on INRIA dataset: (a) image, (b) ground truth, (c) Adapter, (d) LoRA.

the model pre-trained on the remote sensing dataset should be able to improve the performance on downstream remote sensing tasks. But in fact, the effect of remote sensing image pre-training has obvious task discrepancy (Wang et al., 2022c). For the segmentation task, the performance of pre-training based on the remote sensing scene classification dataset is not as good as that pre-trained on ImageNet. How to design a pre-training method suitable to eliminate this task discrepancy is of great significance for improving the remote sensing semantic segmentation tasks (Sun et al., 2022b; Wang et al., 2023e), including building extraction. Additionally, accounting for the unique characteristics of remote sensing data during the training of the foundation model is crucial for its performance in downstream tasks (Hong et al., 2024).

7.3.3. Novel learning paradigm combining multiple multi-source multi-level labeled large-scale datasets

As is known to all, DL is a typical data-driven algorithm. And with the advent of foundation vision models, large-scale datasets become more and more important. For instance, more than 1B masks on 11M images are used to train SAM (Kirillov et al., 2023). To build a powerful agent that can be applied in the actual complex urban environment, an ideal large-scale dataset should simultaneously contain a variety of multi-source data (including HSI, LiDAR, SAR, street-level panorama, *etc.*) from various scenarios and annotations containing fine-grained information (such as building function, height, age, and *etc.*) as much as possible. However, constructing such an ideal dataset is extremely time-consuming and expensive even with the assistance of foundation vision models. Considering the existing large-scale dataset, UBC (Huang et al., 2022b, 2023) provides fine-grained annotations, OmniCity (Li et al., 2022b) contains multi-level and multi-view images, and many other datasets provide abundant multi-modal images, a reasonable alternative solution is to design a novel learning paradigm that combines multi-modal learning, multi-task learning and cross-domain learning. The implementation of this complex architecture requires the joint efforts of scholars from different fields in the future.

8. Conclusion

Building extraction from remote sensing data is a practical and challenging task. With the progress of DL and the development of sensor technology, lots of related studies have come forth. In this survey, a comprehensive and systematic review of recent DL-based building extraction methods is conducted. In particular, an entry-level guidance is presented at first. Then the related works organized by a novel taxonomy are introduced. The proposed taxonomy is from: (1) representation and learning-oriented perspectives and (2) input and output-oriented perspectives. Moreover, the extensive publicly available datasets, products and performance comparisons of recent SOTA methods are summarized. Last but not least, the outlook for future directions is provided from three aspects: practical application oriented algorithm design, joint analysis of multi-source data, foundation vision models and large-scale training.

In conclusion, this survey aims to provide a rough outline of technology development for newcomers and valuable references for researchers. As DL methods have become the mainstream approach to deal with building extraction, it is hoped that this work can offer some insights.

CRediT authorship contribution statement

Yuan Yuan: Writing – review & editing, Validation, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Xiaofeng Shi:** Writing – review & editing, Writing – original draft, Investigation, Data curation, Conceptualization. **Junyu Gao:** Writing – review & editing, Validation, Supervision, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the National Key Research and Development Program of China under Grant 2022ZD0160400 and in part by the Natural Science Foundation of China under Grant 62306241.

Data availability

Data will be made available on request.

References

Alshehhi, R., Marpu, P.R., Woon, W.L., Mura, M.D., 2017. Simultaneous extraction of roads and buildings in remote sensing imagery with convolutional neural networks. ISPRS J. Photogramm. Remote Sens. 130, 139–149.
Audebert, N., Le Saux, B., Lefèvre, S., 2018. Beyond RGB: Very high resolution urban remote sensing with multimodal deep networks. ISPRS J. Photogramm. Remote Sens. 140, 20–32.
Awrangjeb, M., Zhang, C., Fraser, C.S., 2013. Automatic extraction of building roofs using LIDAR data and multispectral imagery. ISPRS J. Photogramm. Remote Sens. 83, 1–18.
Badrinarayanan, V., Kendall, A., Cipolla, R., 2017. SegNet: A deep convolutional encoder-decoder architecture for image segmentation. IEEE Trans. Pattern Anal. Mach. Intell. 39 (12), 2481–2495, arXiv:1511.00561.
Bakirman, T., Komurcu, I., Sertel, E., 2022. Comparative analysis of deep learning based building extraction methods with the new VHR Istanbul dataset. Expert Syst. Appl. 202, 117346.
Bischke, B., Helber, P., Folz, J., Borth, D., Dengel, A., 2019. Multi-task learning for segmentation of building footprints with deep neural networks. In: Proc. ICIP. pp. 1480–1484.

Boo, G., Darin, E., Leasure, D.R., Dooley, C.A., Chamberlain, H.R., Lázár, A.N., Tschirhart, K., Sinai, C., Hoff, N.A., Fuller, T., Musene, K., Batumbo, A., Rimoin, A.W., Tatem, A.J., 2022. High-resolution population estimation using household survey data and building footprints. Nature Commun. 13 (1), 1330.
Bradbury, K., Brigman, B., Collins, L., Johnson, T., Lin, S., Newell, R., Park, S., Suresh, S., Wiesner, H., Xi, Y., 2016. Aerial imagery object identification dataset for building and road detection, and building height estimation. figshare.
Cao, H., Wang, Y., Chen, J., Jiang, D., Zhang, X., Tian, Q., Wang, M., 2021. Swin-unet: Unet-like pure transformer for medical image segmentation. arXiv preprint.
Caye Daudt, R., Le Saux, B., Boulch, A., Gousseau, Y., 2019. Multitask learning for large-scale semantic change detection. Comput. Vis. Image Und. 187, 102783.
Chan, S., Wang, Y., Lei, Y., Cheng, X., Chen, Z., Wu, W., 2023. Asymmetric cascade fusion network for building extraction. IEEE Trans. Geosci. Remote Sens. 61, 1–18.
Chen, J., Bai, X., 2023. Learning to "segment anything" in thermal infrared images through knowledge distillation with a large scale dataset SATIR. arXiv preprint.
Chen, H., Cheng, L., Zhuang, Q., Zhang, K., Li, N., Liu, L., Duan, Z., 2022a. Structure-aware weakly supervised network for building extraction from remote sensing images. IEEE Trans. Geosci. Remote Sens..
Chen, J., He, P., Zhu, J., Guo, Y., Sun, G., Deng, M., Li, H., 2023a. Memory-contrastive unsupervised domain adaptation for building extraction of high-resolution remote sensing imagery. IEEE Trans. Geosci. Remote Sens. 61, 1–15.
Chen, J., Jiang, Y., Luo, L., Gong, W., 2022b. ASF-net: Adaptive screening feature network for building footprint extraction from remote-sensing images. IEEE Trans. Geosci. Remote Sens. 60, 1–13.
Chen, W., Jiang, Z., Wang, Z., Cui, K., Qian, X., 2019. Collaborative global-local networks for memory-efficient segmentation of ultra-high resolution images. In: Proc. CVPR. pp. 8924–8933.
Chen, K., Liu, C., Chen, H., Zhang, H., Li, W., Zou, Z., Shi, Z., 2023b. Rsprompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model. arXiv preprint.
Chen, S., Ogawa, Y., Zhao, C., Sekimoto, Y., 2023c. Large-scale individual building extraction from open-source satellite imagery via super-resolution-based instance segmentation approach. ISPRS J. Photogramm. Remote Sens. 195, 129–152.
Chen, B., Pan, Z., Yang, J., Long, H., 2024a. ACMFNet: Attention-based cross-modal fusion network for building extraction of remote sensing images. IEEE Trans. Geosci. Remote Sens. 62, 1–14.
Chen, L.-C., Papandreou, G., Kokkinos, I., Murphy, K., Yuille, A.L., 2014. Semantic image segmentation with deep convolutional nets and fully connected CRFs. arXiv preprint.
Chen, L.C., Papandreou, G., Kokkinos, I., Murphy, K., Yuille, A.L., 2018a. DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs. IEEE Trans. Pattern Anal. Mach. Intell. 40 (4), 834–848, arXiv:1606.00915.
Chen, L.-C., Papandreou, G., Schroff, F., Adam, H., 2017. Rethinking atrous convolution for semantic image segmentation. arXiv preprint.
Chen, J., Qiu, X., Ding, C., Wu, Y., 2022c. CVCMFF net: Complex-valued convolutional and multifeature fusion network for building semantic segmentation of InSAR images. IEEE Trans. Geosci. Remote Sens. 60, 1–14.
Chen, X., Qiu, C., Guo, W., Yu, A., Tong, X., Schmitt, M., 2022d. Multiscale feature learning by transformer for building extraction from satellite images. IEEE Geosci. Remote Sens. Lett. 19, 1–5.
Chen, S., Shi, W., Zhou, M., Zhang, M., Xuan, Z., 2022e. CGSANet: A contour-guided and local structure-aware encoder–decoder network for accurate building extraction from very high-resolution remote sensing imagery. IEEE J. Sel. Topics Appl. Earth Observations Remote Sens. 15, 1526–1542.
Chen, Y., Tang, L., Yang, X., Bilal, M., Li, Q., 2020. Object-based multi modal convolution neural networks for building extraction using panchromatic and multispectral imagery. Neurocomputing 386, 136–146.
Chen, L.-C., Yang, Y., Wang, J., Xu, W., Yuille, A.L., 2016. Attention to scale: Scale-aware semantic image segmentation. In: Proc. CVPR. pp. 3640–3649.
Chen, T., Zhu, L., Ding, C., Cao, R., Wang, Y., Li, Z., Sun, L., Mao, P., Zang, Y., 2023d. SAM fails to segment anything? – SAM-adapter: Adapting SAM in underperformed scenes: Camouflage, shadow, medical image segmentation, and more. arXiv preprint.
Chen, J., Zhu, J., He, P., Guo, Y., Hong, L., Yang, Y., Deng, M., Sun, G., 2024b. Unsupervised domain adaptation for building extraction of high-resolution remote sensing imagery based on decoupling style and semantic features. IEEE Trans. Geosci. Remote Sens. 62, 1–17.
Chen, L.-C., Zhu, Y., Papandreou, G., Schroff, F., Adam, H., 2018b. Encoder-decoder with atrous separable convolution for semantic image segmentation. In: Proc. ECCV. pp. 801–818.
Chen, K., Zou, Z., Shi, Z., 2021. Building extraction from remote sensing images with sparse token transformers. Remote Sens. 13 (21).
Cheng, D., Liao, R., Fidler, S., Urtasun, R., 2019. DARNet: Deep active ray network for building segmentation. In: Proc. CVPR.
Cheng, G., Xie, X., Han, J., Guo, L., Xia, G.-S., 2020. Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities. IEEE J. Sel. Topics Appl. Earth Observations Remote Sens. 13, 3735–3756.
Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., Bengio, Y., 2014. Learning phrase representations using RNN encoder-decoder for statistical machine translation. In: Proc. EMNLR. pp. 1724–1734.
Ciresan, D., Giusti, A., Gambardella, L., Schmidhuber, J., 2012. Deep neural networks segment neuronal membranes in electron microscopy images. vol. 25.
Claudio, P., Ronny, H., Gemine, V., Kaiqiang, C., Zhiyuan, Y., Deke, T., Hai, H., Michael, S., Xian, S., 0000. 2023 IEEE GRSS data fusion contest: Large-scale fine-grained building classification for semantic urban reconstruction.
Dai, X., Xia, M., Weng, L., Hu, K., Lin, H., Qian, M., 2023. Multiscale location attention network for building and water segmentation of remote sensing image. IEEE Trans. Geosci. Remote Sens. 61, 1–19.
Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., Fei-Fei, L., 2009. ImageNet: A large-scale hierarchical image database. In: Proc. CVPR. pp. 248–255.
Deng, S., Wu, S., Bian, A., Zhang, J., Di, B., Nienkötter, A., Deng, T., Feng, T., 2023. Scattered Mountainous Area building extraction from an open satellite imagery dataset. IEEE Geosci. Remote Sens. Lett. 20, 1–5.
Ding, L., Tang, H., Liu, Y., Shi, Y., Zhu, X.X., Bruzzone, L., 2022. Adversarial shape learning for building extraction in VHR remote sensing images. IEEE Trans. Image Process. 31, 678–690.
Dixit, M., Chaurasia, K., Kumar Mishra, V., 2021. Dilated-ResUnet: A novel deep learning architecture for building extraction from medium resolution multi-spectral satellite imagery. Expert Syst. Appl. 184, 115530.
Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., Houlsby, N., 2021. An image is worth 16x16 words: Transformers for image recognition at scale. In: Proc. ICLR.
Ebel, P., Xu, Y., Schmitt, M., Zhu, X.X., 2022. SEN12MS-CR-TS: A remote-sensing data set for multimodal multitemporal cloud removal. IEEE Trans. Geosci. Remote Sens. 60, 1–14.
Etten, A.V., Lindenbaum, D., Bacastow, T.M., 2018. SpaceNet: A remote sensing dataset and challenge series. arXiv preprint.
Fang, F., Zheng, D., Li, S., Liu, Y., Zeng, L., Zhang, J., Wan, B., 2022. Improved pseudomasks generation for weakly supervised building extraction from high-resolution remote sensing imagery. IEEE J. Sel. Topics Appl. Earth Observations Remote Sens. 15, 1629–1642.
Farabet, C., Couprie, C., Najman, L., LeCun, Y., 2013. Learning hierarchical features for scene labeling. IEEE Trans. Pattern Anal. Mach. Intell. 35 (8), 1915–1929.
Fu, J., Liu, J., Tian, H., Li, Y., Bao, Y., Fang, Z., Lu, H., 2019. Dual attention network for scene segmentation. In: Proc. CVPR. pp. 3141–3149.
Fu, W., Xie, K., Fang, L., 2024. Complementarity-aware local–global feature fusion network for building extraction in remote sensing images. IEEE Trans. Geosci. Remote Sens. 62, 1–13.
Fuentes Reyes, M., Xie, Y., Yuan, X., d'Angelo, P., Kurz, F., Cerra, D., Tian, J., 2023. A 2D/3D multimodal data simulation approach with applications on urban semantic segmentation, building extraction and change detection. ISPRS J. Photogramm. Remote Sens. 205, 74–97.
Ghamisi, P., Rasti, B., Yokoya, N., Wang, Q., Hofle, B., Bruzzone, L., Bovolo, F., Chi, M., Anders, K., Gloaguen, R., Atkinson, P.M., Benediktsson, J.A., 2019. Multisource and multitemporal data fusion in remote sensing: A comprehensive review of the state of the art. IEEE Geosci. Remote Sens. Mag. 7 (1), 6–39.
Ghanea, M., Moallem, P., Momeni, M., 2016. Building extraction from high-resolution satellite images in urban areas: Recent methods and strategies against significant challenges. Int. J. Remote Sens. 37 (21), 5234–5248.
Girard, N., Smirnov, D., Solomon, J., Tarabalka, Y., 2021. Polygonal building extraction by frame field learning. In: Proc. CVPR. pp. 5891–5900.
Gong, M., Liu, T., Zhang, M., Zhang, Q., Lu, D., Zheng, H., Jiang, F., 2023. Context–content collaborative network for building extraction from high-resolution imagery. Knowl-Based. Syst. 263, 110283.
Grinias, I., Panagiotakis, C., Tziritas, G., 2016. MRF-based segmentation and unsupervised classification for building and road detection in peri-urban areas of high-resolution satellite images. ISPRS J. Photogramm. Remote Sens. 122, 145–166.
Guo, H., Du, B., Zhang, L., Su, X., 2022a. A coarse-to-fine boundary refinement network for building footprint extraction from remote sensing imagery. ISPRS J. Photogramm. Remote Sens. 183, 240–252.
Guo, H., Gao, J., Yuan, Y., 2024. Balanced density regression network for remote sensing object counting. IEEE Trans. Geosci. Remote Sens. 62, 1–13.
Guo, S., Liu, L., Gan, Z., Wang, Y., Zhang, W., Wang, C., Jiang, G., Zhang, W., Yi, R., Ma, L., Xu, K., 2022b. ISDNet: Integrating shallow and deep networks for efficient ultra-high resolution segmentation. In: Proc. CVPR. pp. 4361–4370.
Guo, H., Shi, Q., Du, B., Zhang, L., Wang, D., Ding, H., 2021a. Scene-driven multitask parallel attention network for building extraction in high-resolution remote sensing images. IEEE Trans. Geosci. Remote Sens. 59 (5), 4287–4306.
Guo, H., Shi, Q., Marinoni, A., Du, B., Zhang, L., 2021b. Deep building footprint update network: A semi-supervised method for updating existing building footprint from bi-temporal remote sensing images. Remote Sens. Environ. 264, 112589.
Guo, H., Su, X., Wu, C., Du, B., Zhang, L., 2023. Decoupling semantic and edge representations for building footprint extraction from remote sensing images. IEEE Trans. Geosci. Remote Sens. 61, 1–16.
Guo, Z., Wu, G., Song, X., Yuan, W., Chen, Q., Zhang, H., Shi, X., Xu, M., Xu, Y., Shibasaki, R., Shao, X., 2019. Super-resolution integrated building semantic segmentation for multi-source remote sensing imagery. IEEE Access 7, 99381–99397.

Gur, S., Shaharabany, T., Wolf, L., 2020. End to end trainable active contours via differentiable rendering. In: Proc. ICLR.
Hänsch, R., Arndt, J., Lunga, D., Gibb, M., Pedelose, T., Boedihardjo, A., Petrie, D., Bacastow, T.M., 2022. SpaceNet 8 - the detection of flooded roads and buildings. In: Proc. CVPRW. pp. 1472–1480.
Hatamizadeh, A., Sengupta, D., Terzopoulos, D., 2020. End-to-end trainable deep active contour models for automated image segmentation: Delineating buildings in aerial imagery. In: Proc. ECCV. pp. 730–746.
He, K., Gkioxari, G., Dollár, P., Girshick, R., 2017. Mask R-CNN. In: Proc. ICCV. pp. 2961–2969.
He, L., Shan, J., Aliaga, D., 2023. Generative building feature estimation from satellite images. IEEE Trans. Geosci. Remote Sens. 61, 1–13.
He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition. In: Proc. CVPR. pp. 770–778.
Hong, D., Zhang, B., Li, X., Li, Y., Li, C., Yao, J., Yokoya, N., Li, H., Ghamisi, P., Jia, X., Plaza, A., Gamba, P., Benediktsson, J.A., Chanussot, J., 2024. SpectralGPT: Spectral remote sensing foundation model. IEEE Trans. Pattern Anal. Mach. Intell. 46 (8), 5227–5244.
Hosseinpour, H., Samadzadegan, F., Javan, F.D., 2022. CMGFNet: A deep cross-modal gated fusion network for building extraction from very high-resolution remote sensing images. ISPRS J. Photogramm. Remote Sens. 184, 96–115.
Hou, S., Zhang, G., Cui, H., Li, X., Chen, Y., Li, H., Wang, H., Ma, X., 2023. Stable prototype-guided single-temporal supervised learning for change detection and extraction of building. IEEE Trans. Geosci. Remote Sens. 61, 1–22.
Houlsby, N., Giurgiu, A., Jastrzebski, S., Morrone, B., De Laroussilhe, Q., Gesmundo, A., Attariyan, M., Gelly, S., 2019. Parameter-efficient transfer learning for NLP. In: Proc. ICML, 97, pp. 2790–2799.
Hu, J., Shen, L., Sun, G., 2018. Squeeze-and-excitation networks. In: Proc. CVPR. pp. 7132–7141.
Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., 2022. LoRA: Low-rank adaptation of large language models. In: Proc. ICLR.
Hu, Y., Wang, Z., Huang, Z., Liu, Y., 2023a. PolyBuilding: Polygon transformer for building extraction. ISPRS J. Photogramm. Remote Sens. 199, 15–27.
Hu, A., Wu, L., Chen, S., Xu, Y., Wang, H., Xie, Z., 2023b. Boundary shape-preserving model for building mapping from high-resolution remote sensing images. IEEE Trans. Geosci. Remote Sens. 61, 1–17.
Hu, A., Wu, L., Xu, Y., Xie, Z., 2024. SANET: A shape-aware building footprints extraction method in remote sensing images by integrating Fourier shape descriptors. IEEE Trans. Geosci. Remote Sens. 62, 1–15.
Huang, X., Chen, K., Tang, D., Liu, C., Ren, L., Sun, Z., Hänsch, R., Schmitt, M., Sun, X., Huang, H., Mayer, H., 2023. Urban building classification (UBC) V2—A benchmark for global building detection and fine-grained classification from satellite imagery. IEEE Trans. Geosci. Remote Sens. 61, 1–16.
Huang, H., Chen, Y., Wang, R., 2022a. A lightweight network for building extraction from remote sensing images. IEEE Trans. Geosci. Remote Sens. 60, 1–12.
Huang, Z., Cheng, G., Wang, H., Li, H., Shi, L., Pan, C., 2016. Building extraction from multi-source remote sensing images via deep deconvolution neural networks. In: Proc. IGARSS. pp. 1835–1838.
Huang, X., Li, J., Yang, J., Zhang, Z., Li, D., Liu, X., 2021. 30 m global impervious surface area dynamics and urban expansion pattern observed by landsat satellites: From 1972 to 2019. Sci. China Earth Sci. 64, 1922–1933.
Huang, X., Ren, L., Liu, C., Wang, Y., Yu, H., Schmitt, M., Hänsch, R., Sun, X., Huang, H., Mayer, H., 2022b. Urban building classification (UBC) - a dataset for individual building detection and classification from satellite imagery. In: Proc. CVPRW. pp. 1413–1421.
Huang, W., Tang, H., Xu, P., 2022c. OEC-RNN: Object-oriented delineation of rooftops with edges and corners using the recurrent neural network from the aerial images. IEEE Trans. Geosci. Remote Sens. 60, 1–12.
Huang, X., Zhang, Z., Li, J., 2024. China's first sub-meter building footprints derived by deep learning. Remote Sens. Environ. 311, 114274.
Illarionova, S., Nesteruk, S., Shadrin, D., Ignatiev, V., Pukalchik, M., Oseledets, I., 2021. Object-based augmentation for building semantic segmentation: Ventura and santa rosa case study. In: Proc. ICCVW. pp. 1659–1668.
Ji, S., Wei, S., Lu, M., 2019. Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set. IEEE Trans. Geosci. Remote Sens. 57 (1), 574–586.
Ji, D., Zhao, F., Lu, H., 2023. Guided patch-grouping wavelet transformer with spatial congruence for ultra-high resolution segmentation. In: Proc. IJCAI.
Jing, Y., Zhang, T., Liu, Z., Hou, Y., Sun, C., 2023. Swin-ResUNet+: An edge enhancement module for road extraction from remote sensing images. Comput. Vis. Image Und. 237, 103807.
Jung, H., Choi, H.-S., Kang, M., 2022. Boundary enhancement semantic segmentation for building extraction from remote sensed image. IEEE Trans. Geosci. Remote Sens. 60, 1–12.
Kaiser, P., Wegner, J.D., Lucchi, A., Jaggi, M., Hofmann, T., Schindler, K., 2017. Learning aerial image segmentation from online maps. IEEE Trans. Geosci. Remote Sens. 55 (11), 6054–6068.
Kang, J., Fernandez-Beltran, R., Sun, X., Ni, J., Plaza, A., 2022a. Deep learning-based building footprint extraction with missing annotations. IEEE Geosci. Remote Sens. Lett. 19, 1–5.
Kang, J., Körner, M., Wang, Y., Taubenböck, H., Zhu, X.X., 2018. Building instance classification using street view images. ISPRS J. Photogramm. Remote Sens. 145, 44–59.
Kang, J., Wang, Z., Zhu, R., Sun, X., Fernandez-Beltran, R., Plaza, A., 2021. Picoco: Pixelwise contrast and consistency learning for semisupervised building footprint segmentation. IEEE J. Sel. Topics Appl. Earth Observations Remote Sens. 14, 10548–10559.
Kang, J., Wang, Z., Zhu, R., Xia, J., Sun, X., Fernandez-Beltran, R., Plaza, A., 2022b. DisOptNet: Distilling semantic knowledge from optical images for weather-independent building segmentation. IEEE Trans. Geosci. Remote Sens. 60, 1–15.
Kass, M., Witkin, A., Terzopoulos, D., 1988. Snakes: Active contour models. IJCV 1 (4), 321–331.
Ke, L., Danelljan, M., Li, X., Tai, Y.-W., Tang, C.-K., Yu, F., 2022. Mask transfiner for high-quality instance segmentation. In: Proc. CVPR. pp. 4412–4421.
Keisler, R., Skillman, S.W., Gonnabathula, S., Poehnelt, J., Rudelis, X., Warren, M.S., 2019. Visual search over billions of aerial and satellite images. Comput. Vis. Image Und. 187, 102790.
Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A.C., Lo, W.-Y., Dollár, P., Girshick, R., 2023. Segment anything. arXiv preprint.
Krizhevsky, A., Sutskever, I., Hinton, G.E., 2012. Imagenet classification with deep convolutional neural networks. In: Proc. NeurIPS. pp. 1097–1105.
Lee, K., Kim, J.H., Lee, H., Park, J., Choi, J.P., Hwang, J.Y., 2022. Boundary-oriented binary building segmentation model with two scheme learning for aerial images. IEEE Trans. Geosci. Remote Sens. 60, 1–17.
Lee, J.-H., Kim, C., Sull, S., 2021a. Weakly supervised segmentation of small buildings with point labels. In: Proc. ICCV. pp. 7406–7415.
Lee, K., Lee, H., Hwang, J.Y., 2021b. Self-mutating network for domain adaptive segmentation in aerial images. In: Proc. ICCV. pp. 7068–7077.
Li, E., Femiani, J., Xu, S., Zhang, X., Wonka, P., 2015. Robust rooftop extraction from visible band images using higher order CRF. IEEE Trans. Geosci. Remote Sens. 53 (8), 4483–4495.
Li, J., He, W., Cao, W., Zhang, L., Zhang, H., 2024a. UANet: An uncertainty-aware network for building extraction from remote sensing images. IEEE Trans. Geosci. Remote Sens. 62, 1–13.
Li, W., He, C., Fang, J., Zheng, J., Fu, H., Yu, L., 2019. Semantic segmentation-based building footprint extraction using very high-resolution satellite images and multi-source GIS data. Remote Sens. 11 (4).
Li, Y., Hong, D., Li, C., Yao, J., Chanussot, J., 2024b. HD-net: High-resolution decoupled network for building footprint extraction via deeply supervised body and boundary decomposition. ISPRS J. Photogramm. Remote Sens. 209, 51–65.
Li, J., Huang, X., Tu, L., Zhang, T., Wang, L., 2022a. A review of building detection from very high resolution optical remote sensing images. GISci. Remote Sens. 59 (1), 1199–1225.
Li, M., Lafarge, F., Marlet, R., 2020. Approximating shapes in images with low-complexity polygons. In: Proc. CVPR. pp. 8633–8641.
Li, W., Lai, Y., Xu, L., Xiangli, Y., Yu, J., He, C., Xia, G.-S., Lin, D., 2022b. OmniCity: Omnipotent city understanding with multi-level and multi-view images. arXiv preprint.
Li, W., Meng, L., Wang, J., He, C., Xia, G.-S., Lin, D., 2021a. 3D building reconstruction from monocular remote sensing images. In: Proc. ICCV. pp. 12548–12557.
Li, Q., Mou, L., Hua, Y., Shi, Y., Zhu, X.X., 2022c. Building footprint generation through convolutional neural networks with attraction field representation. IEEE Trans. Geosci. Remote Sens. 60, 1–17.
Li, Q., Mou, L., Hua, Y., Shi, Y., Zhu, X.X., 2022d. CrossGeoNet: A framework for building footprint generation of label-Scarce Geographical Regions. Int. J. Appl. Earth. Obs. 111, 102824.
Li, Q., Mou, L., Sun, Y., Hua, Y., Shi, Y., Zhu, X.X., 2024c. A review of building extraction from remote sensing imagery: Geometrical structures and semantic attributes. IEEE Trans. Geosci. Remote Sens. 62, 1–15.
Li, Q., Shi, Y., Zhu, X.X., 2022e. Semi-supervised building footprint generation with feature and output consistency training. IEEE Trans. Geosci. Remote Sens. 60, 1–17.
Li, Q., Yang, W., Liu, W., Yu, Y., He, S., 2021b. From contexts to locality: Ultra-high resolution image segmentation via locality-aware contextual correlation. In: Proc. ICCV. pp. 7252–7261.
Li, X., Zhang, G., Cui, H., Hou, S., Chen, Y., Li, Z., Li, H., Wang, H., 2023a. Progressive fusion learning: A multimodal joint segmentation framework for building extraction from optical and SAR images. ISPRS J. Photogramm. Remote Sens. 195, 178–191.
Li, W., Zhao, W., Yu, J., Zheng, J., He, C., Fu, H., Lin, D., 2023b. Joint semantic–geometric learning for polygonal building segmentation from high-resolution remote sensing images. ISPRS J. Photogramm. Remote Sens. 201, 26–37.
Li, W., Zhao, W., Zhong, H., He, C., Lin, D., 2021c. Joint semantic-geometric learning for polygonal building segmentation. In: Proc. AAAI. pp. 1958–1965.
Lin, H., Hao, M., Luo, W., Yu, H., Zheng, N., 2023. BEARNet: A novel buildings edge-aware refined network for building extraction from high-resolution remote sensing images. IEEE Geosci. Remote Sens. Lett. 20, 1–5.
Lin, J., Jing, W., Song, H., Chen, G., 2019. Esfnet: Efficient network for building extraction from high-resolution aerial images. IEEE Access 7, 54285–54294.

Liu, X., Chen, Y., Wang, C., Tan, K., Li, J., 2023a. A lightweight building instance extraction method based on adaptive optimization of mask contour. Int. J. Appl. Earth. Obs. 122, 103420.

Liu, J., Gong, M., Qin, K., Zhang, P., 2018. A deep convolutional coupling network for change detection based on heterogeneous optical and radar images. IEEE Trans. Neural Netw. Learn. Syst. 29 (3), 545–559.

Liu, K., Huang, J., Xu, M., Perc, M., Li, X., 2021a. Density saliency for clustered building detection and population capacity estimation. Neurocomputing 458, 127–140.

Liu, Y., Li, Q., Yuan, Y., Du, Q., Wang, Q., 2022a. ABNet: Adaptive balanced network for multiscale object detection in remote sensing imagery. IEEE Trans. Geosci. Remote Sens. 60, 1–14.

Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B., 2021b. Swin transformer: Hierarchical vision transformer using shifted windows. In: Proc. ICCV. pp. 10012–10022.

Liu, Z., Shi, Q., Ou, J., 2022b. LCS: A collaborative optimization framework of vector extraction and semantic segmentation for building extraction. IEEE Trans. Geosci. Remote Sens. 60, 1–15.

Liu, Z., Tang, H., 2023. Learning sparse geometric features for building segmentation from low-resolution remote-sensing images. Remote Sens. 15 (7).

Liu, Z., Tang, H., Feng, L., Lyu, S., 2023b. China Building Rooftop Area: the first multi-annual (2016–2021) and high-resolution (2.5 m) building rooftop area dataset in China derived with super-resolution segmentation from sentinel-2 imagery. Earth Syst. Sci. Data 15 (8), 3547–3572.

Liu, G., Yuan, Y., Zhang, Y., Dong, Y., Li, X., 2022c. Style transformation-based spatial-spectral feature learning for unsupervised change detection. IEEE Trans. Geosci. Remote Sens. 60, 1–15.

Liu, Y., Zhao, Z., Zhang, S., Huang, L., 2022d. Multiregion scale-aware network for building extraction from high-resolution remote sensing images. IEEE Trans. Geosci. Remote Sens. 60, 1–10.

Lu, X., Jiao, L., Liu, Q., Li, L., Liu, F., Liu, X., Yang, Y., 2023. Trident cooperation network for building extraction and height estimation. In: Proc. IGARSS. pp. 762–765.

Luo, H., Feng, X., Du, B., Zhang, Y., 2024. A multimodal feature fusion network for building extraction with very high-resolution remote sensing image and LiDAR data. IEEE Trans. Geosci. Remote Sens. 62, 1–19.

Luo, M., Ji, S., Wei, S., 2023. A diverse large-scale building dataset and a novel plug-and-play domain generalization method for building extraction. IEEE J. Sel. Topics Appl. Earth Observations Remote Sens. 16, 4122–4138.

Luo, L., Li, P., Yan, X., 2021. Deep learning-based building extraction from remote sensing images: A comprehensive review. Energies 14 (23).

Maggiori, E., Tarabalka, Y., Charpiat, G., Alliez, P., 2017a. Can semantic labeling methods generalize to any city? The inria aerial image labeling benchmark. In: Proc. IGARSS. pp. 3226–3229.

Maggiori, E., Tarabalka, Y., Charpiat, G., Alliez, P., 2017b. Convolutional neural networks for large-scale remote-sensing image classification. IEEE Trans. Geosci. Remote Sens. 55 (2), 645–657.

Mao, Y., Chen, K., Zhao, L., Chen, W., Tang, D., Liu, W., Wang, Z., Diao, W., Sun, X., Fu, K., 2023a. Elevation estimation-driven building 3-D reconstruction from single-view remote sensing imagery. IEEE Trans. Geosci. Remote Sens. 61, 1–18.

Mao, Y., Sun, X., Huang, X., Chen, K., 2023b. Light: Joint individual building extraction and height estimation from satellite images through a unified multitask learning network. In: Proc. IGARSS. pp. 5320–5323.

Marconcini, M., Metz-Marconcini, A., Üreyen, S., Palacios-Lopez, D., Hanke, W., Bachofer, F., Zeidler, J., Esch, T., Gorelick, N., Kakarla, A., et al., 2020. Outlining where humans live, the world settlement footprint 2015. Sci. Data 7 (1), 242.

Marcos, D., Tuia, D., Kellenberger, B., Zhang, L., Bai, M., Liao, R., Urtasun, R., 2018. Learning deep structured active contours end-to-end. In: Proc. CVPR. pp. 8877–8885.

Mayer, H., 1999. Automatic object extraction from aerial imagery—A survey focusing on buildings. Comput. Vis. Image Und. 74 (2), 138–149.

Milojevic-Dupont, N., Wagner, F., Nachtigall, F., Hu, J., Brüser, G.B., Zumwald, M., Biljecki, F., Heeren, N., Kaack, L.H., Pichler, P.-P., et al., 2023. EUBUCCO v0. 1: European building stock characteristics in a common and open database for 200+ million individual buildings. Sci. Data 10 (1), 147.

Mnih, V., 2013. Machine Learning for Aerial Image Labeling. University of Toronto (Canada).

Mongus, D., Lukač, N., Žalik, B., 2014. Ground and building extraction from LiDAR data based on differential morphological profiles and locally fitted surfaces. ISPRS J. Photogramm. Remote Sens. 93, 145–156.

Na, Y., Kim, J.H., Lee, K., Park, J., Hwang, J.Y., Choi, J.P., 2021. Domain adaptive transfer attack-based segmentation networks for building extraction from aerial images. IEEE Trans. Geosci. Remote Sens. 59 (6), 5171–5182.

Nauata, N., Furukawa, Y., 2020. Vectorizing world buildings: Planar graph reconstruction by primitive detection and relationship inference. In: Proc. ECCV. pp. 711–726.

Peng, X., Wei, Y., Deng, A., Wang, D., Hu, D., 2022. Balanced multimodal learning via on-the-fly gradient modulation. In: Proc. CVPR. pp. 8238–8247.

Prathap, G., Afanasyev, I., 2018. Deep learning approach for building detection in satellite multispectral imagery. In: Proc. Int. Conf. Intell. Syst.. pp. 461–465.

Qi, Z., Chen, H., Liu, C., Shi, Z., Zou, Z., 2023. Implicit ray transformers for multiview remote sensing image segmentation. IEEE Trans. Geosci. Remote Sens. 61, 1–15.

Qiu, C., Li, H., Guo, W., Chen, X., Yu, A., Tong, X., Schmitt, M., 2022. Transferring transformer-based models for cross-area building extraction from remote sensing images. IEEE J. Sel. Topics Appl. Earth Observations Remote Sens. 15, 4104–4116.

Ran, S., Gao, X., Yang, Y., Li, S., Zhang, G., Wang, P., 2021. Building multi-feature fusion refined network for building extraction from high-resolution remote sensing images. Remote Sens. 13 (14), 2794.

Ronneberger, O., Fischer, P., Brox, T., 2015. U-net: Convolutional networks for biomedical image segmentation. In: Proc. MICCAI, vol. 9351, pp. 234–241.

Roscher, R., Volpi, M., Mallet, C., Drees, L., Wegner, J.D., 2020. SemCity toulouse: A benchmark for building instance segmentation in satellite images. In: Proc. ISPRS APRSSIS, vol. 5, pp. 109–116.

Rottensteiner, F., Sohn, G., Gerke, M., Wegner, J.D., 2014. ISPRS semantic labeling contest.

Sabour, S., Frosst, N., Hinton, G.E., 2017. Dynamic routing between capsules. In: Proc. NeurIPS, vol. 30.

Shaloni, Dixit, M., Agarwal, S., Gupta, P., 2020. Building extraction from remote sensing images: A survey. In: Proc. ICACCCN. pp. 966–971.

Shelhamer, E., Long, J., Darrell, T., 2017. Fully convolutional networks for semantic segmentation. IEEE Trans. Pattern Anal. Mach. Intell. 39 (4), 640–651, arXiv:1411.4038.

Shermeyer, J., Hogan, D., Brown, J., Van Etten, A., Weir, N., Pacifici, F., Hansch, R., Bastidas, A., Soenen, S., Bacastow, T., Lewis, R., 2020. SpaceNet 6: Multi-sensor all weather mapping dataset. In: Proc. CVPRW. pp. 196–197.

Shi, X., Gao, J., Yuan, Y., 2024a. Enhancing unimodal features matters: A multimodal framework for building extraction. IEEE Trans. Geosci. Remote Sens. 62, 1–13.

Shi, Q., Zhu, J., Liu, Z., Guo, H., Gao, S., Liu, M., Liu, Z., Liu, X., 2024b. The last puzzle of global building footprints—Mapping 280 million buildings in east Asia based on VHR images. J. Remote Sens. 4, 0138.

Shin, Y.-H., Son, K.-W., Lee, D.-C., 2022. Semantic segmentation and building extraction from airborne LiDAR data with multiple return using PointNet++. Appl. Sci. 12 (4).

Simonyan, K., Zisserman, A., 2015. Very deep convolutional networks for large-scale image recognition. In: Proc. ICLR. arXiv:1409.1556.

Sirmacek, B., Unsalan, C., 2009. Urban-area and building detection using SIFT keypoints and graph theory. IEEE Trans. Geosci. Remote Sens. 47 (4), 1156–1167.

Soliman, A., Chen, Y., Luo, S., Makharov, R., Kindratenko, V., 2022. Weakly supervised segmentation of buildings in digital elevation models. IEEE Geosci. Remote Sens. Lett. 19, 1–5.

Song, S., Zhang, Y., Yuan, Y., 2024. Iterative edge enhancing framework for building change detection. IEEE Geosci. Remote Sens. Lett. 21, 1–5.

Strudel, R., Garcia, R., Laptev, I., Schmid, C., 2021. Segmenter: Transformer for semantic segmentation. In: Proc. ICCV. pp. 7262–7272.

Sun, Y., Hua, Y., Mou, L., Zhu, X.X., 2022a. CG-net: Conditional GIS-aware network for individual building segmentation in VHR SAR images. IEEE Trans. Geosci. Remote Sens. 60, 1–15.

Sun, X., Huang, X., Mao, Y., Sheng, T., Li, J., Wang, Z., Lu, X., Ma, X., Tang, D., Chen, K., 2024. GABLE: A first fine-grained 3D building model of China on a national scale from very high resolution satellite imagery. Remote Sens. Environ. 305, 114057.

Sun, W., Liu, Z., Zhang, Y., Zhong, Y., Barnes, N., 2023. An alternative to wsss? An empirical study of the segment anything model (SAM) on weakly-supervised semantic segmentation problems. arXiv preprint.

Sun, X., Wang, P., Lu, W., Zhu, Z., Lu, X., He, Q., Li, J., Rong, X., Yang, Z., Chang, H., He, Q., Yang, G., Wang, R., Lu, J., Fu, K., 2022b. RingMo: A remote sensing foundation model with masked image modeling. IEEE Trans. Geosci. Remote Sens..

Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., Rabinovich, A., 2015. Going deeper with convolutions. In: Proc. CVPR. pp. 1–9.

Tian, Q., Zhao, Y., Li, Y., Chen, J., Chen, X., Qin, K., 2022. Multiscale building extraction with refined attention pyramid networks. IEEE Geosci. Remote Sens. Lett. 19, 1–5.

Tomljenovic, I., Höfle, B., Tiede, D., Blaschke, T., 2015. Building extraction from airborne laser scanning data: An analysis of the state of the art. Remote Sens. 7 (4), 3826–3862.

Uss, M.L., Vozel, B., Lukin, V.V., Chehdi, K., 2016. Multimodal remote sensing image registration with accuracy estimation at local and global scales. IEEE Trans. Geosci. Remote Sens. 54 (11), 6587–6605.

Van Etten, A., Hogan, D., Manso, J.M., Shermeyer, J., Weir, N., Lewis, R., 2021. The multi-temporal urban development SpaceNet dataset. In: Proc. CVPR. pp. 6398–6407.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I.,

Vats, A., Völgyes, D., Vermeer, M., Pedersen, M., Raja, K., Fantin, D.S.M., Alexander Hay, J., 2024. Terrain-informed self-supervised learning: Enhancing building footprint extraction from LiDAR data with limited annotations. IEEE Trans. Geosci. Remote Sens. 62, 1–10.

Wang, Y., Chen, S., Zhang, R., Xu, F., Huang, P., Yang, W., 2024a. Cross-resolution distillation for building extraction in medium-resolution satellite imagery. In: Proc. IGARSS. pp. 8122–8126.

Wang, L., Fang, S., Meng, X., Li, R., 2022a. Building extraction with vision transformer. IEEE Trans. Geosci. Remote Sens. 60, 1–11.
Wang, X., Kong, T., Shen, C., Jiang, Y., Li, L., 2020. SOLO: Segmenting objects by locations. In: Proc. ECCV. pp. 649–665.
Wang, J., Meng, L., Li, W., Yang, W., Yu, L., Xia, G.-S., 2022b. Learning to extract building footprints from off-nadir aerial images. IEEE Trans. Pattern Anal. Mach. Intell. 45 (1), 1294–1301.
Wang, M., Su, L., Yan, C., Xu, S., Yuan, P., Jiang, X., Zhang, B., 2024b. Rsbuilding: Toward general remote sensing image building extraction and change detection with foundation model. IEEE Trans. Geosci. Remote Sens. 62, 1–17.
Wang, Y., Wang, Z., Cheng, P., Zeng, X., Wang, H., Sun, X., Fu, K., 2023a. DCM: A distributed collaborative training method for the remote sensing image classification. IEEE Trans. Geosci. Remote Sens. 61, 1–17.
Wang, J., Yan, X., Shen, L., Lan, T., Gong, X., Li, Z., 2023b. Scale-invariant multi-level context aggregation network for weakly supervised building extraction. Remote Sens. 15 (5).
Wang, S., Zang, Q., Zhao, D., Fang, C., Quan, D., Wan, Y., Guo, Y., Jiao, L., 2023c. Select, purify, and exchange: A multisource unsupervised domain adaptation method for building extraction. IEEE Trans. Neural Netw. Learn. Syst..
Wang, D., Zhang, J., Du, B., Tao, D., Zhang, L., 2023d. Scaling-up remote sensing segmentation dataset with segment anything model. arXiv preprint.
Wang, D., Zhang, J., Du, B., Xia, G.-S., Tao, D., 2022c. An empirical study of remote sensing pretraining. arXiv preprint.
Wang, D., Zhang, Q., Xu, Y., Zhang, J., Du, B., Tao, D., Zhang, L., 2023e. Advancing plain vision transformer toward remote sensing foundation model. IEEE Trans. Geosci. Remote Sens. 61, 1–15.
Wang, J., Zheng, Z., Ma, A., Lu, X., Zhong, Y., 2021a. LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation. In: NeurIPS.
Wang, Z., Zhou, Y., Wang, F., Wang, S., Qin, G., Zou, W., Wang, Z., Liu, S., Zhu, J., 2023f. Fine-grained building extraction with multispectral remote sensing imagery using the deep model. IEEE Trans. Geosci. Remote Sens. 61, 1–13.
Wang, Y., Zorzi, S., Bittner, K., 2021b. Machine-learned 3D building vectorization from satellite imagery. In: Proc. CVPRW. pp. 1072–1081.
Wangiyana, S., Samczyński, P., Gromek, A., 2022. Data augmentation for building footprint segmentation in SAR images: An empirical study. Remote Sens. 14 (9).
Web, 2017. Dstl satellite imagery feature detection. URL https://www.kaggle.com/c/dstl-satellite-imagery-feature-detection/.
Web, 2018a. 2018 open AI tanzania building footprint segmentation challenge. URL https://competitions.codalab.org/competitions/20100.
Web, 2018b. Mapping challenge. URL https://www.aicrowd.com/challenges/mapping-challenge.
Web, 2021. Google open buildings. URL https://sites.research.google/gr/open-buildings.
Web, 2022a. Flops counter for convolutional networks in pytorch framework. URL https://github.com/sovrasov/flops-counter.pytorch.
Web, 2022b. Microsoft global ML building footprints. URL https://github.com/microsoft/GlobalMLBuildingFootprints.
Web, 2023a. Google temporal open buildings 2.5d temporal dataset. URL https://sites.research.google/gr/open-buildings/temporal.
Web, 2023b. Open street maps. URL https://www.openstreetmap.org/.
Wei, S., Ji, S., Lu, M., 2020. Toward automatic building footprint delineation from aerial images using CNN and regularization. IEEE Trans. Geosci. Remote Sens. 58 (3), 2178–2189.
Wei, S., Zhang, T., Ji, S., Luo, M., Gong, J., 2023. BuildMapper: A fully learnable framework for vectorized building contour extraction. ISPRS J. Photogramm. Remote Sens. 197, 87–104.
Weir, N., Lindenbaum, D., Bastidas, A., Etten, A.V., McPherson, S., Shermeyer, J., Kumar, V., Tang, H., 2019. SpaceNet MVOI: A multi-view overhead imagery dataset. In: Proc. ICCV.
Welling, M., Kipf, T.N., 2016. Semi-supervised classification with graph convolutional networks. In: Proc. ICLR.
Workman, S., Rafique, M.U., Blanton, H., Nathan, J., 2022. Revisiting near/remote sensing with geospatial attention. In: Proc. CVPR. pp. 1778–1787.
Wu, Y., Xu, L., Chen, Y., Wong, A., Clausi, D.A., 2022. TAL: Topography-aware multi-resolution fusion learning for enhanced building footprint extraction. IEEE Geosci. Remote Sens. Lett. 19, 1–5.
Xia, G.-S., Huang, J., Xue, N., Lu, Q., Zhu, X., 2019. GeoSay: A geometric saliency for extracting buildings in remote sensing images. Comput. Vis. Image Und. 186, 37–47.
Xia, J., Yokoya, N., Baier, G., 2022. DML: Differ-modality learning for building semantic segmentation. IEEE Trans. Geosci. Remote Sens. 60, 1–14.
Xia, L., Zhang, X., Zhang, J., Yang, H., Chen, T., 2021. Building extraction from very-high-resolution remote sensing images using semi-supervised semantic edge detection. Remote Sens. 13 (11), 2187.
Xie, Y., Feng, D., Chen, H., Liu, Z., Mao, W., Zhu, J., Hu, Y., Baik, S.W., 2022. Damaged building detection from post-earthquake remote sensing imagery considering heterogeneity characteristics. IEEE Trans. Geosci. Remote Sens. 60, 1–17.
Xie, Y., Tian, J., Zhu, X.X., 2023. A co-learning method to utilize optical images and photogrammetric point clouds for building extraction. Int. J. Appl. Earth. Obs. 116, 103165.
Xie, E., Wang, W., Yu, Z., Anandkumar, A., Alvarez, J.M., Luo, P., 2021. SegFormer: Simple and efficient design for semantic segmentation with transformers. In: Proc. NeurIPS, 34, pp. 12077–12090.
Xing, H., Xiang, J., Xiong, L., Wen, Q., Liu, Q., Wang, Y., 2024. A sparse sharing multitask framework for building footprint extraction from remote sensing imagery following the dual lottery ticket hypothesis. IEEE Trans. Geosci. Remote Sens. 62, 1–16.
Xu, Y., Du, B., Zhang, L., Cerra, D., Pato, M., Carmona, E., Prasad, S., Yokoya, N., Hänsch, R., Le Saux, B., 2019. Advanced multi-sensor optical remote sensing for urban land use and land cover classification: Outcome of the 2018 IEEE grss data fusion contest. IEEE J. Sel. Topics Appl. Earth Observations Remote Sens. 12 (6), 1709–1724.
Xu, L., Li, Y., Xu, J., Guo, L., 2022a. Gated spatial memory and centroid-aware network for building instance extraction. IEEE Trans. Geosci. Remote Sens. 60, 1–14.
Xu, L., Li, Y., Xu, J., Zhang, Y., Guo, L., 2023. BCTNet: Bi-branch cross-fusion transformer for building footprint extraction. IEEE Trans. Geosci. Remote Sens. 61, 1–14.
Xu, P., Tang, H., Ge, J., Feng, L., 2021. Espc_nasunet: An end-to-end super-resolution semantic segmentation network for mapping buildings from remote sensing images. IEEE J. Sel. Topics Appl. Earth Observations Remote Sens. 14, 5421–5435.
Xu, Y., Wu, L., Xie, Z., Chen, Z., 2018. Building extraction in very high resolution remote sensing imagery using deep learning and guided filters. Remote Sens. 10 (1).
Xu, Z., Xu, C., Cui, Z., Zheng, X., Yang, J., 2022b. Cvnet: Contour vibration network for building extraction. In: Proc. CVPR. pp. 1383–1391.
Xu, B., Xu, J., Xue, N., Xia, G.-S., 2022c. Accurate polygonal mapping of buildings in satellite imagery. arXiv preprint.
Xu, H., Zhu, P., Luo, X., Xie, T., Zhang, L., 2022d. Extracting buildings from remote sensing images using a multitask encoder-decoder network with boundary refinement. Remote Sens. 14 (3), 564.
Yan, X., Shen, L., Wang, J., Deng, X., Li, Z., 2022. MSG-SR-net: A weakly supervised network integrating multiscale generation and superpixel refinement for building extraction from high-resolution remotely sensed imageries. IEEE J. Sel. Topics Appl. Earth Observations Remote Sens. 15, 1012–1023.
Yang, J., Matsushita, B., Zhang, H., 2023a. Improving building rooftop segmentation accuracy through the optimization of unet basic elements and image foreground-background balance. ISPRS J. Photogramm. Remote Sens. 201, 123–137.
Yang, B., Zhang, M., Zhang, Z., Zhang, Z., Hu, X., 2023b. TopDiG: Class-agnostic topological directional graph extraction from remote sensing images. In: Proc. CVPR. pp. 1265–1274.
Yi, Y., Zhang, Z., Zhang, W., Zhang, C., Li, W., Zhao, T., 2019. Semantic segmentation of urban buildings from VHR remote sensing imagery using a deep convolutional neural network. Remote Sens. 11 (15).
Yu, Y., Liu, C., Gao, J., Jin, S., Jiang, X., Jiang, M., Zhang, H., Zhang, Y., 2022. Building extraction from remote sensing imagery with a high-resolution capsule network. IEEE Geosci. Remote Sens. Lett. 19, 1–5.
Yu, H., Yang, Z., Tan, L., Wang, Y., Sun, W., Sun, M., Tang, Y., 2018. Methods and datasets on semantic segmentation: A review. Neurocomputing 304, 82–103.
Yuan, J., 2018. Learning building extraction in aerial scenes with convolutional networks. IEEE Trans. Pattern Anal. Mach. Intell. 40 (11), 2793–2798.
Yuan, Q., Mohd Shafri, H.Z., 2022. Multi-modal feature fusion network with adaptive center point detector for building instance extraction. Remote Sens. 14 (19).
Yuan, X., Shi, J., Gu, L., 2021. A review of deep learning methods for semantic segmentation of remote sensing imagery. Expert Syst. Appl. 169, 114417–114430.
Zang, N., Cao, Y., Wang, Y., Huang, B., Zhang, L., Mathiopoulos, P.T., 2021. Land-use mapping for high-spatial resolution remote sensing image via deep learning: A review. IEEE J. Sel. Topics Appl. Earth Observations Remote Sens. 14, 5372–5391.
Zhang, S., Cao, Y., Sui, B., 2023a. DF-mask R-CNN: Direction field-based optimized instance segmentation network for building instance extraction. IEEE Geosci. Remote Sens. Lett. 20, 1–5.
Zhang, W., Dai, X., Chu, Q., Liao, Z., Hu, S., Zhang, J., Li, G., Ding, H., Jin, F., 2024a. FFSSNet: Fast fine semantic segmentation network for GF-3 SAR images in building areas. IEEE J. Sel. Topics Appl. Earth Observations Remote Sens. 17, 11260–11273.
Zhang, L., Dong, R., Yuan, S., Li, W., Zheng, J., Fu, H., 2021a. Making low-resolution satellite images reborn: A deep learning approach for super-resolution building extraction. Remote Sens. 13 (15).
Zhang, M., Du, Y., Hu, Z., Liu, Q., Wang, Y., 2023b. BISVP: Building footprint extraction via bidirectional serialized vertex prediction. In: Proc. ICASSP. pp. 1–5.
Zhang, P., Du, P., Lin, C., Wang, X., Li, E., Xue, Z., Bai, X., 2020. A hybrid attention-aware fusion network (HAFNet) for building extraction from high-resolution imagery and LiDAR data. Remote Sens. 12 (22).
Zhang, B., Gao, J., Yuan, Y., 2024b. Center-enhanced video captioning model with multimodal semantic alignment. Neural Netw. 180, 106744.
Zhang, H., Liao, Y., Yang, H., Yang, G., Zhang, L., 2022a. A local-global dual-stream network for building extraction from very-high-resolution remote sensing images. IEEE Trans. Neural Netw. Learn. Syst. 33 (3), 1269–1283.
Zhang, R., Lin, B., Zhang, Q., Zhang, G., 2022b. Homogeneous aggregation convolution for building extraction from remote sensing images. IEEE Geosci. Remote Sens. Lett. 19, 1–5.

Zhang, J., Liu, R., Shi, H., Yang, K., Reiß, S., Peng, K., Fu, H., Wang, K., Stiefelhagen, R., 2023c. Delivering arbitrary-modal semantic segmentation. arXiv preprint.

Zhang, Z., Qian, Z., Zhong, T., Chen, M., Zhang, K., Yang, Y., Zhu, R., Zhang, F., Zhang, H., Zhou, F., et al., 2022c. Vectorized rooftop area data for 90 cities in China. Sci. Data 9 (1), 66.

Zhang, R., Wan, Z., Zhang, Q., Zhang, G., 2023d. DSAT-net: Dual spatial attention transformer for building extraction from aerial images. IEEE Geosci. Remote Sens. Lett. 20, 1–5.

Zhang, N., Wang, G., Wang, J., Chen, H., Liu, W., Chen, L., 2023e. All adder neural networks for on-board remote sensing scene classification. IEEE Trans. Geosci. Remote Sens. 61, 1–16.

Zhang, F., Xu, X., Nauata, N., Furukawa, Y., 2021b. Structured outdoor architecture reconstruction by exploration and classification. In: Proc. ICCV. pp. 12427–12435.

Zhang, R., Zhang, Q., Zhang, G., 2023f. SDSC-UNet: Dual skip connection vit-based U-shaped model for building extraction. IEEE Geosci. Remote Sens. Lett. 20, 1–5.

Zhao, W., Persello, C., Stein, A., 2021. Building outline delineation: from aerial images to polygons with an improved end-to-end learning framework. ISPRS J. Photogramm. Remote Sens. 175, 119–131.

Zhao, W., Persello, C., Stein, A., 2022. Extracting planar roof structures from very high resolution images using graph neural networks. ISPRS J. Photogramm. Remote Sens. 187, 34–45.

Zhao, H., Shi, J., Qi, X., Wang, X., Jia, J., 2017. Pyramid scene parsing network. In: Proc. CVPR. pp. 2881–2890.

Zhao, Y., Sun, G., Zhang, L., Zhang, A., Jia, X., Han, Z., 2023. MSRF-net: Multiscale receptive field network for building detection from remote sensing images. IEEE Trans. Geosci. Remote Sens. 61, 1–14.

Zheng, D., Li, S., Fang, F., Zhang, J., Feng, Y., Wan, B., Liu, Y., 2023. Utilizing bounding box annotations for weakly supervised building extraction from remote-sensing images. IEEE Trans. Geosci. Remote Sens. 61, 1–17.

Zheng, S., Lu, J., Zhao, H., Zhu, X., Luo, Z., Wang, Y., Fu, Y., Feng, J., Xiang, T., Torr, P.H.S., Zhang, L., 2021a. Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers. In: Proc. CVPR. pp. 6881–6890.

Zheng, Z., Ma, A., Zhang, L., Zhong, Y., 2021b. Deep multisensor learning for missing-modality all-weather mapping. ISPRS J. Photogramm. Remote Sens. 174, 254–264.

Zhou, Y., Chen, Z., Wang, B., Li, S., Liu, H., Xu, D., Ma, C., 2022. BOMSC-net: Boundary optimization and multi-scale context awareness based building extraaction from high-resolution remote sensing imagery. IEEE Trans. Geosci. Remote Sens. 60, 1–17.

Zhu, Y., Huang, B., Fan, Y., Usman, M., Chen, H., 2024a. Iterative polygon deformation for building extraction. IEEE Trans. Geosci. Remote Sens. 62, 1–14.

Zhu, Y., Huang, B., Gao, J., Huang, E., Chen, H., 2022. Adaptive polygon generation algorithm for automatic building extraction. IEEE Trans. Geosci. Remote Sens. 60, 1–14.

Zhu, Q., Li, Z., Song, T., Yao, L., Guan, Q., Zhang, L., 2024b. Unrestricted region and scale: Deep self-supervised building mapping framework across different cities from five continents. ISPRS J. Photogramm. Remote Sens. 209, 344–367.

Zhu, Q., Liao, C., Hu, H., Mei, X., Li, H., 2021. MAP-net: Multiple attending path neural network for building footprint extraction from remote sensed imagery. IEEE Trans. Geosci. Remote Sens. 59 (7), 6169–6181.

Ziaja, M., Bosowski, P., Myller, M., Gajoch, G., Gumiela, M., Protich, J., Borda, K., Jayaraman, D., Dividino, R., Nalepa, J., 2021. Benchmarking deep learning for on-board space applications. Remote Sens. 13 (19).

Zorzi, S., Bazrafkan, S., Habenschuss, S., Fraundorfer, F., 2022. PolyWorld: Polygonal building extraction with graph neural networks in satellite images. In: Proc. CVPR. pp. 1848–1857.

Zorzi, S., Bittner, K., Fraundorfer, F., 2021. Machine-learned regularization and polygonization of building segmentation masks. In: Proc. ICPR. pp. 3098–3105.